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FORGED IN FRAGMENTS: POPULATION III STAR FORMATION IN MILKY WAY PROGENITOR HALO

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Abstract

We present a radiation-magnetohydrodynamic (RMHD) cosmological zoom-in simulation of Population III star formation, drawing initial conditions from the FIRE m12f halo at high redshift. Starting from a self-consistent cosmological environment, we follow the collapse of a primordial gas cloud through the formation of an accretion disk that becomes gravitationally unstable within $\sim 3\,\mathrm{kyr}$ of the first protostar's formation. Prominent spiral arms develop and fragment vigorously, producing ~ 60 sink formation events over the $\sim 15\,\mathrm{kyr}$ of post-collapse evolution, with a total stellar mass reaching $> 150\,M_\odot$. The stellar mass grows as $M_* \overset{\propto}{\sim} t^2$, confirming the prediction of accelerating accretion in self-gravitating turbulent collapse, at rates of $\dot{M} \sim 10^{-3}$–$10^{-2}\,M_\odot\,\mathrm{yr}^{-1}$. The disk is initially magnetically supercritical ($\mu \sim 100$–300) but the mass-to-flux ratio decreases by two orders of magnitude over the simulation, with the inner disk approaching and locally crossing the critical value ($\mu \lesssim 1$) by $15\,\mathrm{kyr}$. Magnetic energy grows by ~ 4 orders of magnitude, driven by flux-freezing compression and differential rotation. The star formation efficiency reaches $\sim 80\%$ within the disk and $\sim 50\%$ within a 0.1 pc aperture. Our results demonstrate that cosmologically-realistic initial conditions, which produce asymmetric infall and complex angular momentum structure, lead to vigorous fragmentation and a protostellar population whose properties differ from those found in idealised setups.

1. INTRODUCTION

The first generation of stars, Population III (Population III), formed from metal-free primordial gas in dark-matter minihaloes at redshifts $z \sim 20$–30 (Abel et al. 2002; Bromm et al. 2002). Because molecular hydrogen (H_2) was the only efficient gas coolant at these epochs, the primordial gas could cool to a temperature floor of $T \sim 200$ K at densities $n \sim 10^4\,\mathrm{cm}^{-3}$, before reaching local thermodynamic equilibrium in its ro-vibrational levels (Bromm et al. 2002; Klessen and Glover 2023). The Jeans mass at this point, $M_J \sim 10^3\,M_\odot$, is orders of magnitude larger than that of present-day star-forming cores, leading to the early expectation that Population III stars were predominantly massive and formed in isolation within each minihalo (Abel et al. 2002; Bromm et al. 2002; Yoshida et al. 2008). These early simulations bridged cosmological collapse down to protostellar densities ($n \sim 10^{21}\,\mathrm{cm}^{-3}$), resolving the characteristic stages of primordial collapse: H_2 three-body formation heating at $n \gtrsim 10^8\,\mathrm{cm}^{-3}$, collision-induced emission cooling at $n \gtrsim 10^{14}\,\mathrm{cm}^{-3}$, and a second collapse driven by H_2 dissociation at $n \sim 10^{16}\,\mathrm{cm}^{-3}$ (Yoshida et al. 2008; Wollenberg et al. 2020).

This picture of isolated, monolithic Population III star formation was overturned by simulations that continued to follow the system *after* the initial protostar formed. The infalling gas inevitably carries angular momentum, seeding a self-gravitating accretion disk prone to gravitational instability and fragmentation (Stacy et al. 2010; Clark et al. 2011; Greif et al. 2011). Rather than a single object, a protostellar *cluster* arises: Greif et al. (2011) found 2–5 fragments in five minihaloes within ~ 100 yr of the first protostar; Stacy et al. (2010, 2012) followed systems for $\sim 5\,\mathrm{kyr}$ and found ~ 10 sinks with a primary reaching 20–$40\,M_\odot$. Wollenberg et al. (2020) systematically explored 45 controlled simulations varying initial rotation and turbulence and found 5–20 sinks per halo, with fragmentation strongly stochastic from realization to realization. The protostellar cluster is shaped by competitive accretion (Bonnell et al. 2001), fragmentation-induced starvation (Wollenberg et al. 2020), protostellar mergers, and dynamical ejections, with ejection rates of ~ 25–30% found across multiple studies (Greif et al. 2011; Stacy et al. 2012; Wollenberg et al. 2020).

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The Population III initial mass function (IMF) inferred from these simulations is broadly top-heavy, spanning sub-solar to $\gtrsim 100\,M_{\odot}$, but highly variable from halo to halo. Hirano et al. (2014) surveyed 100 minihaloes and found accretion rates of 10^{-4}–$10^{-2}\,M_{\odot}\,\mathrm{yr}^{-1}$ and final masses of $\sim$ 10–1000 $M_{\odot}$; Susa et al. (2014) found masses spanning 1–$> 100\,M_{\odot}$ across several tens of halos. When radiative feedback is included, accretion onto a single protostar can be halted at $\sim 43\,M_{\odot}$ (Hosokawa et al. 2011), though episodic accretion from a fragmenting disk can produce intermittent feedback and final masses from ~ 10 to several hundred $M_{\odot}$ (Hosokawa et al. 2016; Stacy et al. 2016). The large statistical survey of Hirano et al. (2023) across > 100 cosmological halos confirms that the Population III IMF is not universal: the outcome depends on halo mass, angular momentum, and baryon streaming velocity, and a single simulation cannot be over-generalised.

Turbulence plays a fundamental role in shaping fragmentation and the resulting mass distribution. Chen et al. (2025) demonstrated that cosmological infall into dark-matter potential wells universally drives supersonic turbulence in Population III minihaloes, with Mach numbers $\mathcal{M} \sim 2$–4 that increase with halo mass. Wollenberg et al. (2020) confirmed that the degree of initial turbulence strongly affects the number of fragments and the resulting mass distribution, with higher initial turbulent velocities producing more fragments and more stochastic outcomes. Prieto et al. (2011) showed that turbulent fragmentation in a larger, $3 \times 10^{7}\,M_{\odot}$ halo at $z \sim 11$ produces ~ 25 gravitationally unstable clumps, proposing that infall-driven turbulence may be the dominant fragmentation mechanism when halo masses exceed the canonical $10^{6}\,M_{\odot}$ minihalo scale.

Magnetic fields can further modify Population III star formation, though their net effect remains debated. Even trace cosmological seed fields are amplified during gravitational collapse by flux-freezing and winding, driving the mass-to-flux ratio toward criticality over timescales of a few kyr (Machida et al. 2008; Sharda et al. 2021; Higashi et al. 2024). Idealised MHD simulations suggest that strong fields can suppress disk fragmentation (Hirano and Machida 2022), and cosmological MHD runs by Stacy et al. (2022) found only two secondary sinks. In contrast, Prole et al. (2022) and Tanaka et al. (2024) found that even dynamically significant fields reduce but do not eliminate fragmentation. Sharda and Menon (2025) performed the first 3D radiation-MHD Population III simulations with non-equilibrium chemistry, turbulence, and ionising feedback, finding that magnetic fields are more effective than radiation in limiting protostellar masses at early times, capping the most massive star at $\sim 65\,M_{\odot}$ within 5 kyr (see also Prole et al. 2025). These conflicting results indicate that the outcome depends sensitively on initial field strength, field geometry, resolution, and simulation duration (Klessen and Glover 2023).

While many Population III simulations now use cosmological initial conditions (Greif et al. 2011; Stacy et al. 2016; Hirano et al. 2023; Latif and Khochfar 2022; Meziani et al. 2026), simulations that *simultaneously* combine cosmological ICs, full radiation-MHD, and resolved individual protostellar formation remain few. Meziani et al. (2026) use the FORGE'd in FIRE framework (Hopkins et al. 2023) to study Population III star formation in an ultra-faint dwarf galaxy progenitor, with a DM halo mass of $10^{7}M_{\odot}$ at z=21 and $10^{12}M_{\odot}$ at z=0. A minihalo embedded in a strong overdensity destined to grow to Milky Way mass may experience qualitatively different infall rates, tidal torques, and angular momentum injection compared to one in a quieter environment destined to remain a dwarf galaxy. Therefore, in this paper, we zoom-in to the formation of the first stars in a massive halo $10^{7}M_{\odot}$ that eventually grows to Milky Way mass at z=0.

Whether the environmental difference between a MW-progenitor (Milky Way) and a UFD-progenitor (Ultra Faint Dwarf) drives a qualitatively different outcome is an open question. In this work, we address this question directly. We perform a cosmological radiation-magnetohydrodynamic (RMHD) zoom-in simulation of Population III star formation in the FIRE m12f initial conditions (Hopkins et al. 2018; Wetzel et al. 2023), which in turn were taken from the AGORA project (Kim et al. 2013, 2016). The ICs constitute a halo destined to become a $M_{\rm vir} \sim 10^{12}\,M_{\odot}$ Milky Way analogue by $z = 0$. We use the FORGE'd in FIRE framework (Hopkins et al. 2023; Meziani et al. 2026), combining FIRE-3 (Hopkins et al. 2022) large-scale physics with STARFORGE (Grudić et al. 2021) protostellar physics, and evolve the system for ~ 15 kyr after the first protostar forms, reaching a gas mass resolution of $\Delta m \sim 10^{-3}\,M_{\odot}$. We study the thermodynamic collapse, disk formation and fragmentation, magnetic field amplification, and protostellar mass assembly in this environment, providing the first cosmological RMHD simulation that resolves individual Population III protostellar formation in a MW-progenitor halo.

The rest of the paper is organised as follows. section 2 describes the simulation setup, physics modules, refinement strategy, and analysis methods. section 3 presents the disk morphology, thermodynamic and energetic evolution, disk stability, and magnetic field structure. subsection 3.4 discusses the star formation history, accretion rates, and protostellar mass function. section 4 compares our results to previous Population III simulations. We state our conclusions in section 5.

2. SIMULATION AND METHODS

2.1. *Simulation details*

For this paper we utilise the FORGE'd in FIRE framework of Hopkins et al. (2023), extended to the primordial star-formation regime similar to Meziani et al. (2026). We employ FIRE-3 (Hopkins et al. 2022) physics modules at low resolution and STARFORGE (Grudić et al. 2021) physics to resolve individual protostellar formation. We briefly review these below and refer the reader to Hopkins et al. (2022, 2023); Grudić et al. (2021). Numerical parameters not specified below are identical to those described in the above references.

2.1.1. *Code and gravity solver*

The governing equations solved by the code are the ideal MHD equations in conservative form (Hopkins

2015; Hopkins and Raives 2015):

$$\frac{\partial \mathbf{U}}{\partial t} + \nabla \cdot \mathbf{F} = \mathbf{S}, \tag{1}$$

where the state vector of conserved quantities and the flux tensor are

$$\mathbf{U} = \begin{pmatrix} \rho \\ \rho\mathbf{v} \\ \rho e \\ \mathbf{B} \end{pmatrix}, \qquad \mathbf{F} = \begin{pmatrix} \rho\mathbf{v} \\ \rho\mathbf{v}\otimes\mathbf{v} + P_T\,\mathcal{I} - \mathbf{B}\otimes\mathbf{B} \\ (\rho e + P_T)\mathbf{v} - (\mathbf{v}\cdot\mathbf{B})\mathbf{B} \\ \mathbf{v}\otimes\mathbf{B} - \mathbf{B}\otimes\mathbf{v} \end{pmatrix}. \tag{2}$$

Here ρ is the mass density, $\mathbf{v}$ the velocity, $\mathbf{B}$ the magnetic field (in Gaussian units with 4π absorbed), $e = u + |\mathbf{B}|^2/(2\rho) + |\mathbf{v}|^2/2$ the total specific energy, $P_T = P + |\mathbf{B}|^2/2$ the total (thermal plus magnetic) pressure, and $\mathcal{I}$ the identity tensor. $\mathbf{S}$ includes source terms for self-gravity and the Powell divergence-cleaning scheme (Hopkins and Raives 2015) that maintains $\nabla\cdot\mathbf{B} \approx 0$.

The FORGE'd in FIRE framework is based on GIZMO Meshless Finite Mass (MFM) code (Hopkins 2015). It is a mesh-free Lagrangian Godunov method in which the fluid equations are solved on a set of moving partition-generating points whose masses are conserved exactly (no inter-particle mass flux), while fluxes of momentum, energy, and magnetic flux are exchanged through effective interparticle faces defined by a kernel-weighted partition of unity applied to the particle distribution (Hopkins 2015).

Gravity is solved with the TreePM algorithm (Hopkins 2015), which combines a particle-mesh grid for large-scale forces with an oct-tree for short-range forces (Barnes and Hut 1986), as in Springel (2010). Gas particles use adaptive Lagrangian gravitational softening matching the hydrodynamic kernel radius. Sink/star particles use a fixed Plummer-equivalent softening $\epsilon \approx 20$ AU. Direct N-body summation is used for all sink–sink and sink–gas gravitational interactions, with direct summation applied for sink–sink separations below 1000 AU to accurately resolve close binaries and dynamical ejections (Grudić et al. 2021).

2.1.2. *Initial Conditions*

We use initial conditions (ICs) developed as part of the FIRE collaboration (Hopkins et al. 2018), specifically the m12f halo from the Latte suite of Milky Way-mass zoom-in simulations (Wetzel et al. 2023). The m12f halo has a virial mass $M_{\rm vir} \sim 1$–$2\times10^{12}\,M_\odot$ at $z = 0$ and was selected as an isolated system with a relatively quiescent late-time merger history, making it a Milky Way analogue. At the redshift of first collapse in our simulation ($z \sim 21$), CDM part of the progenitor halo has a mass of $2\times10^7\,M_\odot$, typical of the minihaloes in which Population III stars are expected to form. This value was obtained from running the ROCKSTAR halo finder (Behroozi et al. 2013) on our snapshot.

We adopt pre-generated ICs at $z = 99$ in a $60\,\mathrm{Mpc}/h$ comoving cosmological box. The box contains a zoom-in region of comoving extent $\sim 3\,\mathrm{Mpc}/h$, within which the FIRE galaxy forms at $z = 0$. Only the zoom-in region contains baryonic particles and high-resolution dark matter (DM) particles; the remainder of the volume is populated with low-resolution DM particles. As in the FIRE-2 Latte simulations, we adopt cosmological parameters from WMAP-7 (Komatsu et al. 2011): $h = 0.702$, $\Omega_m = 0.272$, $\Omega_\Lambda = 0.728$, $\Omega_b = 0.0455$, $\sigma_8 = 0.807$, and $n_s = 0.961$

The full simulation snapshot contains $\sim$ 88 million CDM particles and FIRE-resolution gas particles whose smoothing lengths ($\sim$kpc) far exceed the protostellar disk scale. We extract spatial cutouts centred on the most-massive sink particle with a physical radius of 1 kpc (22 ckpc). The cutout sphere retains only the gas, high-resolution DM, and sink/star particles. All analysis presented in this manuscript is of this cutout simulation.

Over the $\sim$ 15 kyr of post-collapse evolution studied here, the sound-crossing time across the cutout radius ($r_{\rm cut} \approx 1$ kpc) is $\sim r_{\rm cut}/c_s \gtrsim 10$ kyr for $T \lesssim 10^4$ K, comparable to or longer than our integration time. The large-scale structure outside the cutout therefore has no causal influence on the disk dynamics on these timescales, and the cutout fully captures the relevant physics.

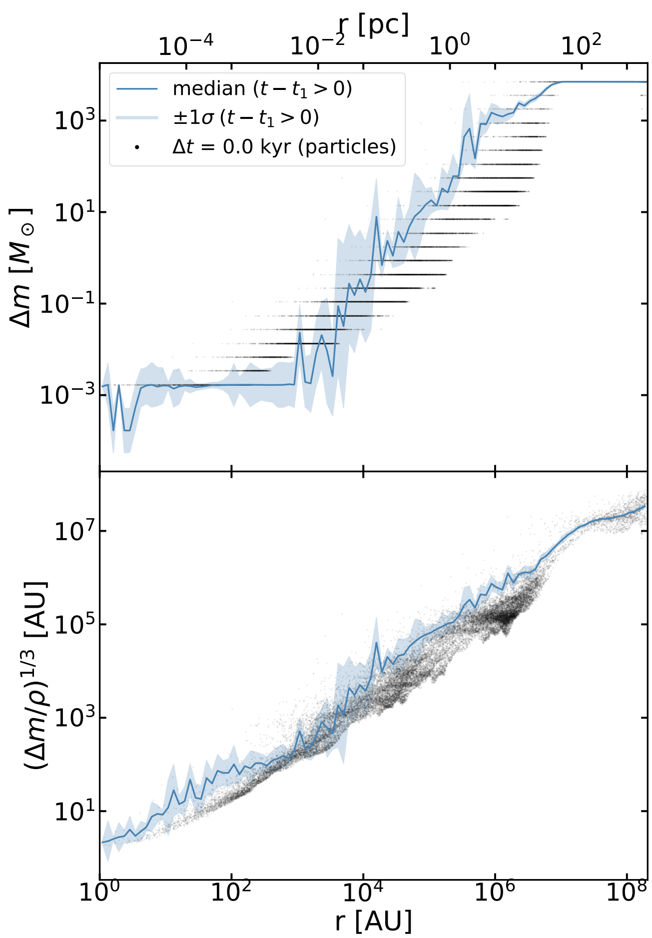

Fig. 1.— Top: Radial profile of gas particle mass (resolution). Scatter plot shows particle distribution at the time of first sink formation; solid line and shaded region correspond to the mean and $\pm1\sigma$ for radial profiles across all snapshots after first sink formation. Horizontal lines correspond to $1/2\Delta m$ of previous resolution region, since our particle splitting algorithm is subdividing parent particles in 2 for each new level of split. Bottom: Same as Top, but for radial profile of equivalent cell size $\Delta x = (\Delta m/\rho)^{1/3}$.

2.1.3. *Refinement strategy*

We first evolve the full cosmological simulation with FIRE-3 physics until first $10^5 M_\odot$ stellar population (SSP) particle forms. We then restart from the preceding snapshot with an additional "hyper-refinement" layer applied globally (Hopkins et al. 2023).

Gas cells are refined (split into pairs of equal-mass daughter cells) whenever their mass exceeds a fraction f_J of the local Jeans mass:

$$\begin{aligned}\Delta m > & f_J M_J \\ &= f_J \frac{\pi}{6}\rho\lambda_J^3 \\ &= f_J \frac{\pi}{6}\rho\left(\frac{\pi c_s^2}{G\rho}\right)^{3/2} \\ &= f_J \frac{\pi^{5/2}}{6}\left(\frac{k_B T}{\mu m_H G}\right)^{3/2}\rho^{-1/2} \\ &= f_J M_0 \left(\frac{T}{10^4\ \mathrm{K}}\right)^{1.5}\left(\frac{n_H}{\mathrm{cm}^{-3}}\right)^{-0.5}\end{aligned} \tag{3}$$

The last line follows from substituting the thermal Jeans length $\lambda_J = (\pi c_s^2/G\rho)^{1/2}$ and $c_s^2 = k_B T/(\mu m_H)$ into $M_J = (\pi/6)\rho\lambda_J^3$, and normalizing at $T_0 = 10^4$ K and $n_0 = 1$ cm^{-3}. We use $f_J = 10^{-2}$, corresponding to ~ 100 resolution elements per Jeans mass and satisfying the Truelove et al. (1997) criterion, since we resolve $100^{1/3} \approx 4.6 > 4$ cells per Jeans length. This is sufficient since MFM does not suffer from artificial fragmentation even while refining (Khullar et al. 2026, in prep.). We also speculate that since MFM is generally less diffusive than AMR, the more stringent criteria of 30 cells per Jeans length needed to resolve effects like magnetic dynamo amplification (Federrath et al. 2011) does not necessarily apply here. Since our simulation produces magnetic amplification by up to 4 orders of magnitude (see subsection 3.3), we leave detailed numerical checks to future work.

Only gas particles get split, DM particles stay at the same resolution. This is a fine approximation because dark matter's potential is smooth at these scales, and splitting of these particles should not change the results of our simulation.

At our highest refinement level, the gas mass resolution is $\Delta m \approx 10^{-3}\,M_\odot$ and the spatial resolution is $\Delta x \approx 2$ AU, sufficient to resolve densities up to $n \sim 10^{15}\,\mathrm{cm}^{-3}$.

Figure 1 shows the radial profile of the length scale of gas particles $\lambda = (m/\rho)^{1/3}$. It exhibits the refinement profile, described in subsubsection 2.1.3 at all times, except the earilest one. Initial profile at $t - t_1 = -0.5$ (0.5 kyrs before the first sink formation) is limited by the refinement scale inside of 100 AU radius (the center was not yet refined to the highest resolution). But by the time of first sink formation, the refinement reaches highest resolution that we expect from our scheme, (around $10^{-3} M_\odot$ or a few AU) so the inner star forming disk region is fully refined.

2.2. *Resolved physics*

2.2.1. *Magnetic fields*

The ideal MHD equations are integrated as described in Hopkins and Raives (2015), using the constrained-gradient method and divergence cleaning (Hopkins 2016) for divergence control, which has been validated for magnetorotational instability dynamics (Hopkins et al. 2023).

Spitzer–Braginskii thermal conduction and viscosity are included via Hopkins (2017). For metal-free gas at the temperatures ($T \sim 200$–10^4 K) and free-electron fractions relevant here, these transport timescales exceed the local dynamical timescale by many orders of magnitude (Meziani et al. 2026), so ideal MHD dominates throughout the simulation.

The initial magnetic field is a trace cosmological seed field of $|B_0| \sim 10^{-15}$ G amplified self-consistently by gravitational compression, differential rotation, and the turbulent dynamo.

2.2.2. *Radiation*

Radiation transport is solved with the M1 moments method (Levermore 1984), as implemented in GIZMO and described in Hopkins et al. (2023); Grudić et al. (2021); Hopkins and Grudić (2019); Hopkins et al. (2020) across five frequency bands: H-ionizing, Lyman-Werner (LW)/photo-electric, Near-UV (NUV), optical–NIR, and an adaptive-wavelength FIR band that tracks the evolving dust emission temperature (Hopkins et al. 2022). Gas, dust, and radiation temperatures are evolved separately with appropriate coupling terms, allowing self-consistent treatment of both optically thin and optically thick regimes. We use a reduced speed of light $\tilde{c} = 10^{-3}c$ following Hopkins et al. (2023).

The Lyman-Werner (LW) band is combined with the photo-electric (PE) heating band into a single radiation frequency bin. This is justified because both LW and PE photons originate from the same protostellar sources and have similar mean free paths at the densities resolved here. H_2 photo-dissociation by LW photons is accounted for through the Γ_{H_2} destruction term in Equation 4.

2.2.3. *Thermochemistry*

Gas thermodynamics and chemistry is solved as in (Hopkins et al. 2022). The following cooling functions are employed and are relevant to our setup: $\Lambda_{\rm H}$ (atomic H cooling); $\Lambda_{\rm HD}$ (HD cooling, Galli and Palla 1998); $\Lambda_{\rm H_2}$ (H_2 cooling); $\Lambda_{\rm H_2}^{\rm thin}$, $\Lambda_{\rm H_2}^{\rm thick}$ (H_2 interactions with e^-, H^+, H_2, H, He in optically thin and thick regimes). We follow non-equilibrium H and He chemical networks. Default FIRE-3 thermochemistry includes cooling due to metal lines, but we do not evolve our simulation past the point of supernova-driven metal enrichment, thus making all metal cooling contributions effectively zero (as expected in the Population III formation regime).

Since we simulate metal-free primordial gas, the dust-to-gas ratio is zero and the dust term in the H_2 formation equation is absent. H_2 formation proceeds exclusively via gas-phase channels: the two-body H^- and H_2^+ pathways at low densities ($n \lesssim 10^8\,\mathrm{cm}^{-3}$) and the three-body reaction ($3\mathrm{H} \rightarrow \mathrm{H_2} + \mathrm{H}$) at high densities.

Molecular fraction of H_2 evolves in every cell according to (eq. 6 from Hopkins et al. 2022), with the dust formation term set to zero:

$$\begin{aligned}\frac{d\bar{n}_{\rm H_2}}{dt} &= \alpha_{\rm GP} C_2 \bar{n}_{\rm HI}\bar{n}_{\rm H} + \alpha_{\rm 3B} C_3 \bar{n}_{\rm HI}\frac{\bar{n}_{\rm H_2}}{8}\left(\bar{n}_{\rm HI} + \frac{\bar{n}_{\rm H_2}}{8}\right) \\ &\quad - \Gamma_{\rm H_2}\bar{n}_{\rm H_2} - \xi_{\rm H_2}\bar{n}_{\rm H_2} - \sum_i \beta_{{\rm H_2},i} C_2 \bar{n}_{\rm H_2}\bar{n}_i\end{aligned} \tag{4}$$

Here $\bar{n}_x$ is volume-averaged number density of x inside of a cell. $C_m = \langle n^m \rangle / \langle n \rangle^m$ is the $m^{\rm th}$-order clumping factor. $\alpha_{\rm GP}$ is the gas-phase formation rate coefficient via the H^- and H_2^+ pathways (Glover and Jappsen 2007; Abel et al. 1997; Galli and Palla 1998). The three-body formation coefficient $\alpha_{\rm 3B}$ (Abel et al. 2002; Palla et al. 1983) becomes the dominant H_2 formation channel for $n \gtrsim 10^8\,{\rm cm^{-3}}$. Γ_{H_2} is the coefficient associated with LW and ionizing photon dissociation. ξ_{H_2} accounts for cosmic-ray-induced ionization and dissociation ($\xi_{H_2} \approx 3\times10^{-17}$ s^{-1}, the Milky Way value; this rate is unconstrained at $z \sim 21$ and primarily affects H_2 formation at $n \lesssim 10^7$ cm^{-3}, with negligible impact at the densities resolved here). The last sum runs over collisional dissociation contributions from e^-, H^+, H_2, HI, HeI, HeII, with rate coefficients $\beta_{H_2,{\rm i}}$ from Glover and Abel (2008).

Thermodynamic energy release from H_2 dissociation ($\sim$ 4.5 eV per molecule) is not explicitly tracked in the gas energy equation. This is acceptable because we do not have enough resolution to reach number densities of $n \approx 10^{16}{\rm cm^{-3}}$, where H_2 dissociation kicks in and becomes a non-negligible thermodynamic variable. (Klessen and Glover 2023) In addition, the protostellar evolution tracks of Offner et al. (2009) self-consistently evolve the stellar radius without requiring the gas to track dissociation energy explicitly.

At $z \sim 21$, the meta-galactic UV background (Chan et al. 2019) is negligible compared to local protostellar radiation resolved in the simulation. H_2 self-shielding against Lyman-Werner photons is included using the shielding function of Gnedin and Draine (2014) (their Eq. 3), applied with a local Sobolev column density $N{\rm H} \sim \rho/|\nabla\rho|$, with turbulent line broadening accounted for following a Burgers velocity spectrum.

The simulation does not include cosmic-ray transport. In our simulation, which covers only $\sim$ 15 kyr of post-collapse evolution and does not include supernovae, cosmic-ray injection from stellar feedback is negligible; the initial cosmic-ray background at $z \sim 21$ contributes subdominantly to H_2 dissociation and ionization relative to the resolved radiation field.

2.2.4. *Stellar formation and evolution*

We resolve the formation of individual protostars in the densest region of our simulation using the STARFORGE (Grudić et al. 2021) formation and evolution modules. A gas particle is converted to a sink (protostar) particle when it meets the full STARFORGE formation criteria: the cell must be (i) locally self-gravitating (virial parameter $\alpha_{\rm vir} < 1$), (ii) Jeans unstable, (iii) in a converging flow ($\nabla \cdot \mathbf{v} < 0$), (iv) experiencing compressive tides in all directions, (v) a local maximum of the density field, and (vi) not within the accretion radius of a pre-existing sink (Grudić et al. 2021).

Once formed, sink particles accrete bound gas whose circularised orbit falls within the sink radius. Each sink evolves along protostellar tracks following Offner et al. (2009), explicitly tracking the stellar radius, luminosity, and effective temperature as a function of the instantaneous mass and accretion rate. Protostellar jets are launched with a mass-loading proportional to the surface accretion rate and a launch velocity comparable to the escape velocity from the protostellar surface (Grudić et al. 2021).

2.3. *Postprocessing methods*

2.3.1. *Disk identification*

To compute velocity dispersion, local Toomre Q, and other disk properties, we identify gas particles belonging to the disk at each snapshot using the following procedure:

1. The disk centre is placed at the most massive sink particle. This is more stable than the mass-weighted centre-of-mass (COM) of all sinks once secondary sinks form at large separations. Before any sinks form, the centre is placed at the densest gas particle within a refinement region.

2. A mass-weighted COM velocity $v_{\rm COM}$ is computed from gas particles within $r < r_{\rm search} = 10^{-5}\,{\rm kpc} \approx 2060\,{\rm AU}$ and subtracted from all particle velocities.

3. The disk normal $\hat{L}$ is computed from the total angular momentum $\mathbf{L} = \sum_i m_i\,(\mathbf{r}_i \times \mathbf{v}_i)$ summed over particles within $r_{\rm search}$, using the COM-subtracted velocities from the previous step. The sign of $\hat{L}$ is fixed so that $L_z > 0$ to prevent frame-to-frame flipping.

4. Particle positions and velocities are projected onto the disk-plane coordinate system ($r_{\rm cyl}$, ϕ, z, v_r, v_ϕ, v_z).

5. Geometric and density cuts are applied: $r_{\rm cyl} < r_{\rm max} = 10^{-5}\,{\rm kpc}$, $|z|/r_{\rm cyl} < 0.3$ (aspect ratio), and $\rho > \rho_{\rm thresh} = 10^{-15}\,{\rm g\,cm^{-3}}$.

6. Kinematic cuts are applied: $v_\phi > 0$ (co-rotating) and $v_\phi/v_K > f_{\rm kep} = 0.3$, where $v_K = \sqrt{GM_{\rm enc}(<r)/r}$ is the Keplerian speed computed from the cumulative enclosed gas mass plus the total sink mass placed at the origin.

7. The kinematically selected set is extended to its bounding cylinder (100th percentile in $r_{\rm cyl}$ and $|z|$) to recover any disk material just outside the strict kinematic boundary.

Particles passing all cuts are labelled as disk members. The disk radius $r_{\rm disk}$ is taken as the 90th-percentile cylindrical radius of disk members.

3. RESULTS

In this section we present results from our cosmological RMHD simulation of Population III star formation. We organize the discussion as follows: subsection 3.1 examines the thermodynamic and energetic evolution during the pre-stellar collapse and early disk phase; subsection 3.2 characterizes the disk structure and gravitational stability; subsection 3.3 describes the magnetic field evolution; and subsection 3.4 covers the star formation history, accretion rates, and protostellar mass function.

Figure 2 presents a multi-scale density projection of the simulation at $t - t_1 \approx 15\,{\rm kyr}$, illustrating the range of scales our simulation resolves after the onset of active star formation.

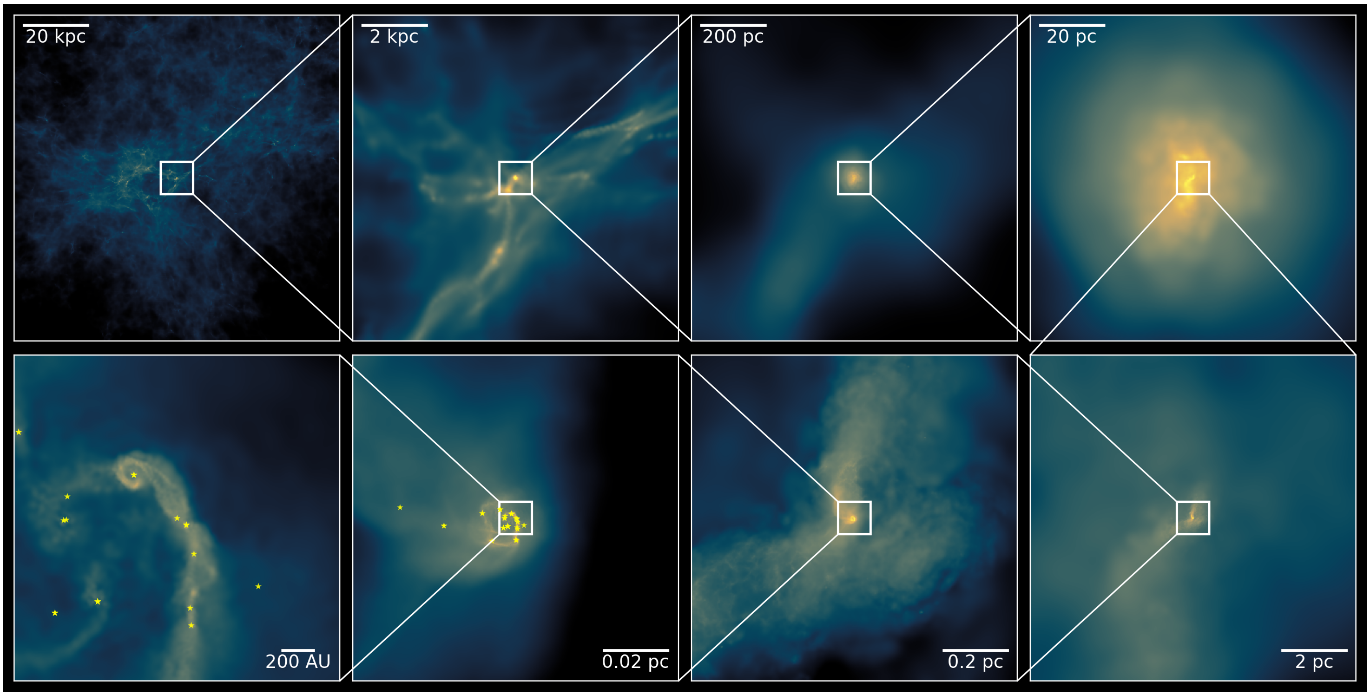

FIG. 2.— Multi-scale density projection of the simulation at $t - t_1 \approx 15\,\mathrm{kyr}$, illustrating the dynamic range from the cosmic web (box size of $\approx 100\,\mathrm{kpc}$) to the center of the protostellar disk (resolving AU scales). White boxes mark the region shown in the next panel. White star symbols mark sink particles.

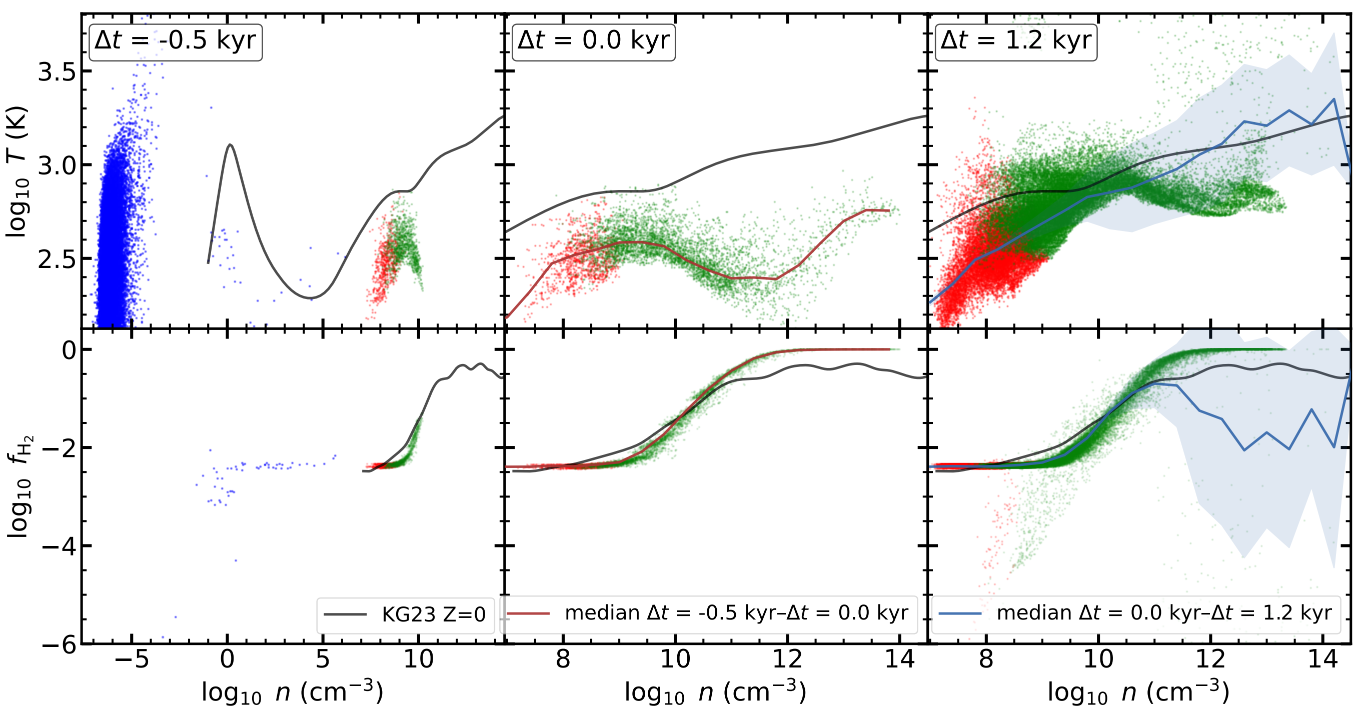

FIG. 3.— Phase diagram (temperature and H_2 fraction vs. number density) at three representative epochs: pre-stellar ($t - t_1 \approx -0.5\,\mathrm{kyr}$, left), first sink formation ($t - t_1 = 0$, middle), and post-first-sink ($t - t_1 = 1.2\,\mathrm{kyr}$, right). Green points correspond to disk-identified particles, red points correspond to points in our cutout simulation that have not been identified as disk, and blue points correspond to points in the full-box simulation right before the cutout was applied (only included in the leftmost plots). $f_{H_2} = 0$ for most of particles in the low-density region (blue particles), thus missing from the bottom-left panel. The dashed curves in the temperature panels show the theoretical T–n tracks for primordial ($Z = 0$) and metal-enriched ($Z = 10^{-4}\,Z_\odot$) gas from Klessen and Glover (2023) (KG23); shaded bands indicate the median $\pm$ standard deviation of the fits over the time intervals $-0.5\,\mathrm{kyr} \leq t - t_1 \leq 0$ (middle) and $0 \leq t - t_1 \leq 1.2\,\mathrm{kyr}$ (right), respectively.

3.1. *Collapse*

3.1.1. *Thermodynamics*

Phase diagrams of temperature and H_2 fraction versus number density are shown in Figure 3 for three representative epochs (pre-stellar, first sink formation, post-first sink formation). Disk particles (identified by the algorithm of subsubsection 2.3.1) are marked in green, particles that are a part of full simulation but not cutout are marked in blue, and particles that are present in cutout simulation but not identified as disk particles are marked in red.

At the pre-stellar epoch ($t - t_1 = -0.5\,$kyr; Figure 3, left panels), the gas occupies a narrow track spanning $n \sim 10^7$–$10^{10}\,\mathrm{cm^{-3}}$ and $T \sim 200$–800 K. This track traces the initial slow collapse, during which H_2 rovibrational cooling brings the gas down to the ~ 200 K temperature floor, followed by gradual compressional heating as H_2 line cooling saturates and becomes optically thick. The H_2 fraction (lower panel) rises modestly from $f_{H_2} \sim 10^{-2.5}$ at $n \sim 10^7\,\mathrm{cm^{-3}}$ to $f_{H_2} \sim 10^{-1.3}$ at $n \sim 10^{10}\,\mathrm{cm^{-3}}$, driven primarily by two-body formation channels (H^- and H_2^+ pathways). Note that increase in H_2 fraction coincides with cooling of the gas, since H_2 is much more efficient coolant compared to atomic H. The low-density gas particles ($n \leq 10^{-5}$ cm^{-3}) in the left column correspond to low-resolution gas in the original zoom-in box before re-simulation (blue markers). Most of those particles have $f_{H_2} = 0$, thus not present in the bottom plot. This part of the simulation domain is outside our region of interest and does not change significantly over our integration time of $\approx 15\,$kyr.

The T–n track at the moment of first sink collapse ($t - t_1 = 0$) follows the $Z = 10^{-4}\,Z_\odot$ metallicity track of Omukai (2000); Klessen and Glover (2023) rather than the zero-metallicity track. We speculate that this slight offset may reflect small departures from chemical equilibrium in the collapsing gas; we do not impose a metallicity floor, so trace metal enrichment from surrounding regions cannot be ruled out as a contributing factor. By the later epoch ($t - t_1 = 1.2\,$kyr), the median of the phase distribution shifts back toward the $Z = 0$ track, suggesting that the gas at this stage is governed more by primordial H_2 chemistry. We defer a systematic one-zone test of this offset to future work.

By $t - t_1 = 0$ (Figure 3, middle panels), the gas has collapsed to much higher densities ($n \sim 10^{13}\,\mathrm{cm^{-3}}$) and the thermal track has extended significantly. At intermediate densities ($n \sim 10^8$–$10^{10}\,\mathrm{cm^{-3}}$), compressional heating outpaces H_2 line cooling, which saturates as the lines become optically thick, and the temperature rises steadily through $T \sim 500$–800 K. The H_2 fraction increases steeply in this regime as three-body formation reactions ($3H \rightarrow H_2 + H$) become efficient at $n \gtrsim 10^8\,\mathrm{cm^{-3}}$, converting the gas to fully molecular ($f_{H_2} \rightarrow 1$) by $n \sim 10^{11}$–$10^{12}\,\mathrm{cm^{-3}}$. Note that unlike in the pre-stellar regime, we see an increase in temperature associated with increase in H_2 fraction. This is a second-order effect: H_2 cools the gas very effectively, reducing thermal pressure, allowing the gas to continue the infall process. This makes the gas opaque to H_2 lines, heating it up. (Ripamonti and Abel 2004)

At $t - t_1 = 1.2\,$kyr (Figure 3, right panels), the phase diagram has broadened substantially. The gas has separated into 2 parts: a continuously-infalling branch with $T < 2000$ K and $n < 10^{10}$ cm^{-3}, and a hot branch at $T \sim 2000$–8000 K from shock-heated and adiabatically compressed gas. The H_2 fraction now shows a bimodality at high densities: gas that has been heated to $T \gtrsim 2000$ K via radiation produced by the first formed sink undergoes collisional H_2 dissociation. We note that while our thermochemical model does not couple H_2 dissociation to thermodynamics, the $\beta_{H_2,i}$ terms in Equation 4 can still drive partial dissociation of H_2.

A low-f_{H_2} population ($f_{H_2} \sim 10^{-3}$–10^{-5}) is produced at $n \sim 10^{10}$–$10^{12}\,\mathrm{cm^{-3}}$, while the lower-density gas keeps following the infall trajectory, driven by 2-body H_2 production pathway that the collapse started with (i.e. left panels). (Klessen and Glover 2023)

At later times, the three-body pathway to H_2 production becomes dominant way of converting atomic hydrogen to molecular in collapsing regions. This results in constantly increasing fraction of molecular hydrogen in the disk with time.

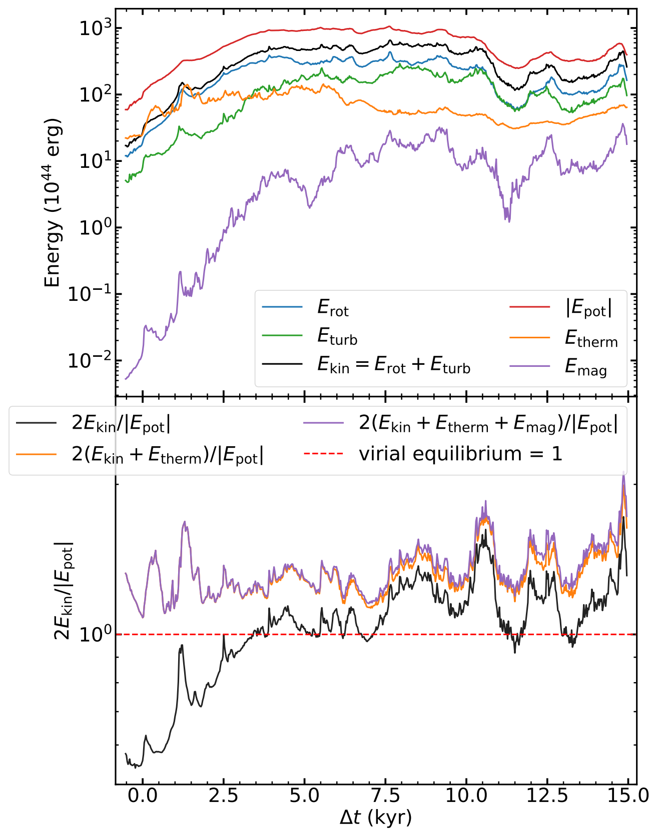

FIG. 4.— Top: Time evolution of the main energy components computed for disk-identified particles: rotational kinetic energy $E_{\rm rot}$ (yellow), turbulent kinetic energy $E_{\rm turb}$ (cyan), total kinetic energy $E_{\rm kin} = E_{\rm rot} + E_{\rm turb}$ (green), gravitational potential energy $|E_{\rm pot}|$ (red), thermal energy $E_{\rm therm}$ (orange), and magnetic energy $E_{\rm mag}$ (magenta).
Bottom: pure kinematic virial ratio $2E_{\rm kin}/|E_{\rm pot}|$ (black); virial ratio with added thermal component (orange), and with added thermal and magnetic components (purple). The dashed red line marks virial equilibrium (= 1). The disk starts sub-virial and reaches equilibrium by $\sim 2\,$kyr. The inclusion of thermal energy reveals that the disk is thermally supported and near virial equilibrium at all times. Magnetic energy provides negligible additional support to the virial ratio.

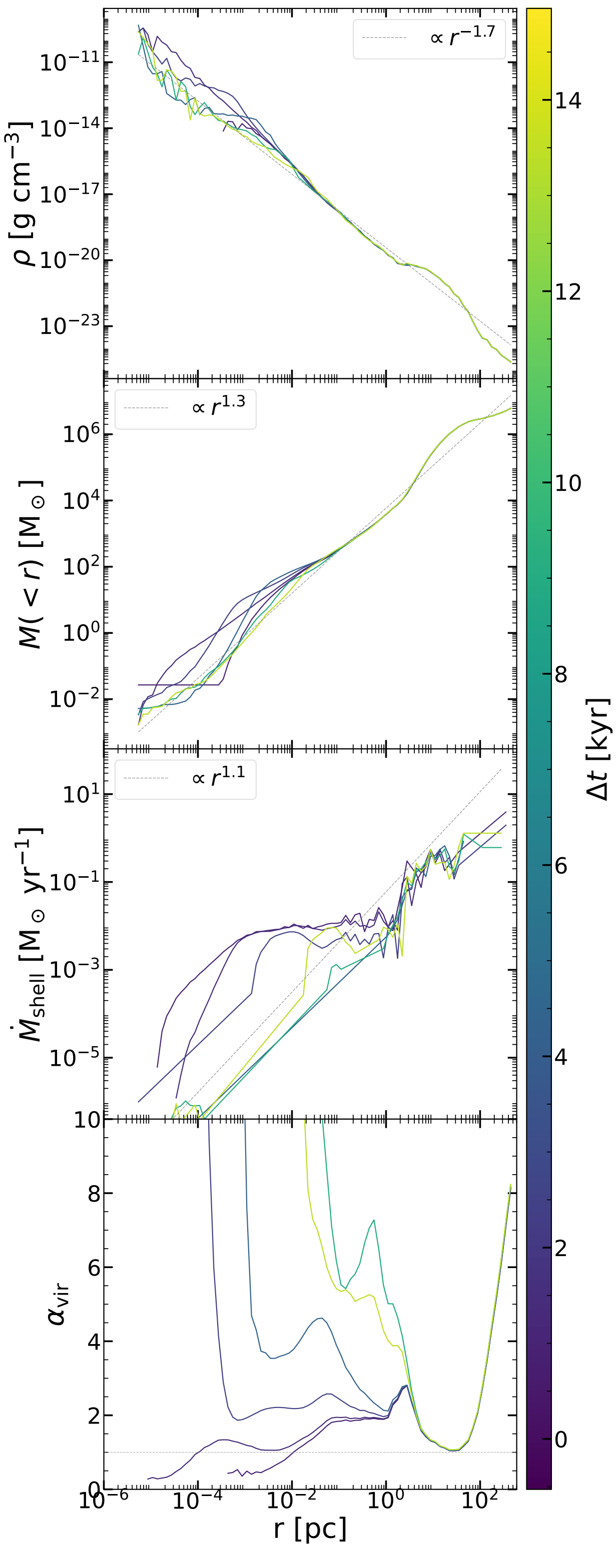

FIG. 5.— First plot: Radial density profiles before the formation of the first sink. The x-axis begins at ~ 1 AU, corresponding to the resolution limit of the simulation. The fit for the region outside of the disk is consistently $\rho \propto r^{-3.5}$ accross all times.
Second plot: Gas mass in a ball vs radius of the ball before the formation of the first sink. The fit is roughly Bottom: rate of change of mass per shell, dM/dt. General trend is that the outer shells gain mass faster. **[TODO: Fix caption]**

3.1.2. *Pre-stellar outside-in collapse*

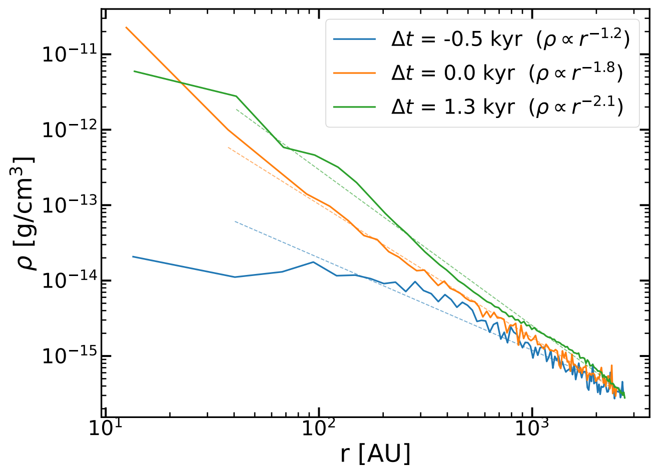

FIG. 6.— Radial density profiles at three different times before the first bulk star formation event at $t - t_1 \approx 1.2$ kyr. The density increases with decreasing radius, and follows a power law relationship that steepens with time.

We characterize the collapse of the gas cloud before the formation of the first sink particle by computing gas mass in spherical shells centered on the densest particle in the disk (which will later correspond to the first sink particle location). We perform this calculation for every snapshot, with a time interval between snapshots of $\Delta t \approx 0.1$ kyr, and show the shell mass profiles in the top panel of Figure 5. The rate of mass change per shell, $\Delta m/\Delta t$, is shown in the bottom panel.

The fact that the outer layers gain mass faster demonstrates an outside-in collapse. This contrasts with the inside-out model of Shu (1977) and aligns with the collapse dynamics of gravitationally unstable Bonnor-Ebert spheres (Ebert 1955; Bonnor 1956; Larson 1969; Penston 1969). Figure 7 confirms this: the infall velocity $(-v_r)$ is supersonic in the envelope outside the disk ($r > 2500$ AU), which is a definitive signature of outside-in dynamics. Within the disk ($r < 2500$ AU), the flow becomes predominantly subsonic. This abrupt deceleration occurs because the infalling gas encounters an accretion shock as it strikes the dense, rotationally supported disk material.

3.2. *Disk evolution*

3.2.1. *Energetics*

In this section, we consider the interplay of the gravitational, kinetic, thermal and magnetic energies for the particles identified in the disk. We compute the following energy components: potential $E_{\rm pot}$, rotational $E_{\rm rot}$, turbulent $E_{\rm turb}$, thermal E_{therm}, and magnetic $E_{\rm mag}$, using Equations 5:

$$E_{\rm pot} = -G \sum_{\substack{i<j \\ i,j\in{\rm disk}}} \frac{m_i\, m_j}{\sqrt{|\mathbf{r}_i - \mathbf{r}_j|^2 + \epsilon^2}} \tag{5a}$$

$$- \; G \sum_{s\in{\rm sinks}} \sum_{i\in{\rm disk}} \frac{m_s\, m_i}{\sqrt{|\mathbf{r}_s - \mathbf{r}_i|^2 + \epsilon^2}},$$

$$E_{\rm rot} = \tfrac{1}{2} \sum_{i\in{\rm disk}} m_i v_{\phi,i}^2, \tag{5b}$$

$$E_{\rm turb} = \tfrac{1}{2} \sum_{i\in{\rm disk}} m_i |\mathbf{v}_i - v_{\phi,i}\hat{\boldsymbol{\phi}} - v_{r,i}\hat{\mathbf{r}}|^2, \tag{5c}$$

$$E_{\rm therm} = \sum_{i\in{\rm disk}} m_i u_i, \tag{5d}$$

$$E_{\rm mag} = \sum_{i\in{\rm disk}} \frac{|\mathbf{B}_i|^2}{8\pi} \frac{m_i}{\rho_i} \tag{5e}$$

where i denotes all gas particles identified as disk members (see subsubsection 2.3.1), m_i is the mass of particle, $\mathbf{v}_i$ is the total velocity vector, $v_{\phi,i}, v_{r,i}$ are bulk velocity profile fits for azimuthal and radial components for each annulus (both are functions of r only), $\mathbf{B}_i$ is the magnetic field vector, ρ_i is the density, u_i is the internal energy.

Figure 4 shows the time evolution of the main energy components computed for disk-identified particles. Throughout the simulation, $|E_{\rm pot}|$ dominates over all other components, confirming that the disk is gravitationally bound. $E_{\rm rot}$ and $E_{\rm turb}$ are the main support terms and track each other to within a factor of ~ 2, with $E_{\rm rot}$ slightly exceeding $E_{\rm turb}$. $E_{\rm therm}$ starts comparable to the kinetic energies at early times but becomes subdominant at late times as the gravitational potential well deepens faster than the gas heats.

The magnetic energy, $E_{\rm mag}$, over the ~ 15 kyr simulation, from $\sim 10^{42}$ erg at $t - t_1 \approx -0.5$ kyr to $\sim 10^{45.5}$ erg at 15 kyr. This growth is driven by a combination of flux-freezing compression during the collapse, field-line winding by differential rotation, and potentially the small-scale turbulent dynamo (Sharda et al. 2021). Despite this rapid amplification, $E_{\rm mag}$ remains below $E_{\rm therm}$ throughout the simulation, though the gap narrows at late times – consistent with the declining mass-to-flux ratios discussed in subsection 3.3.

The bottom panel shows the virial ratio $2E_{\rm kin}/|E_{\rm pot}|$. At early times ($t - t_1 < 0$), the virial ratio is ~ 0.6–0.7, indicating a *sub-virial*, contracting state, i.e. the system has insufficient kinetic energy to balance gravity.

The virial ratio rises as the disk forms and spins up, reaching ~ 1.0 by $t - t_1 \approx 2$ kyr. From 2 kyr onward, the virial ratio oscillates around unity, indicating that the disk has achieved and maintains approximate virial equilibrium through a balance of rotational support, turbulent pressure, and gravitational collapse. A dip in virial ratio after 11 kyr likely corresponds to the slow-down of stellar accretion rate (when less gas is accreted per unit time, less gravitational energy is converted to kinetic energy, reducing the kinetic-to-potential ratio), which can also be observed in Figure 18. The "bounce-back" times also align with the "bounce-back" in the stellar accretion rate values.

The virial ratio never approaches the unbound threshold of 2, confirming that the system remains gravitationally bound throughout.

3.2.2. *Disk morphology and evolution*

?? shows radial profiles of velocity components of gas particles in the disk, as well as turbulent velocity magnitude and dispersion. Radial component v_r is taken with a negative, with positiven direction now corresponding to radially inward, to simplify comparison

The general trend is that v_ϕ, $|\delta v|$, σ_r increase with time everywhere. v_r is negative and thus gas is mostly infalling for the whole simulation, with $|v_r|$ increasing until stabilizing at around 1.3-3.6 kyr (bulk star formation events), and dropping after stars are fully formed. c_s rises until 1.3 kyr, after which it stabilizes. It is clear that the turbulence starts off as sonic or sub-sonic, but becomes greatly supersonic once the sink forms.

Figure 8 shows face-on and edge-on surface density projections at six representative epochs spanning the full ~ 15 kyr evolution after the formation of the first sink (t_1).

At the pre-stellar epoch ($t - t_1 = -0.5$ kyr), the gas is centrally concentrated without significant rotational support, appearing as an elongated, filamentary structure. By the time the first sink forms ($t - t_1 = 0$), a compact central concentration has developed. A coherent disk with recognisable rotational structure emerges within ~ 1 kyr. By $t - t_1 \approx 3.6$ kyr, the disk develops prominent two-armed spiral structure as it becomes gravitationally unstable (subsubsection 3.2.3), with multiple sink particles visible along the spiral arms. By $t - t_1 \approx 9.5$ kyr, the disk hosts ~ 20 visible sink particles embedded in a multi-armed, irregular spiral pattern extending to ~ 2000 AU, with several sinks located far from the centre. t

Radial density profile evolution is shown on Figure 6. It is clear that the density profile is following a steepening power law $\rho \propto r^{-\alpha}$, with $\alpha \approx 1.2$ at the time before the first sink forms, $\alpha \approx 1.8$ at the point of formation of the first sink, and $\alpha \approx 2.1$ by the time of a first bulk star formation event, at the point when the spiral arms have formed.

3.2.3. *Disk stability and fragmentation*

[SK: Can you please rewrite the paper beyond this point with the corrections I suggested for the earlier parts of the paper?]

To characterise non-thermal motions in the disk, we decompose particle velocities into radial and azimuthal components (v_r, v_ϕ) in the disk-plane coordinate system. We subdivide the disk into concentric cylindrical annuli and compute the mass-weighted mean velocity in each annulus, yielding radial profiles $\bar{v}_r(r)$ and $\bar{v}_\phi(r)$.

The residual (streaming-subtracted) velocity for each particle is $\vec{\delta v} = (v_r - \bar{v}_r,\ v_\phi - \bar{v}_\phi,\ v_z)$. The velocity dispersion σ_v is computed as the root-mean-square of $|\vec{\delta v}|$ within each annulus.

The Toomre Q parameter quantifies the gravitational stability of a differentially rotating disk:

$$Q = \frac{c_s \kappa}{\pi G \Sigma}, \tag{6}$$

where c_s is the sound speed, κ the epicyclic frequency, and Σ the surface density.

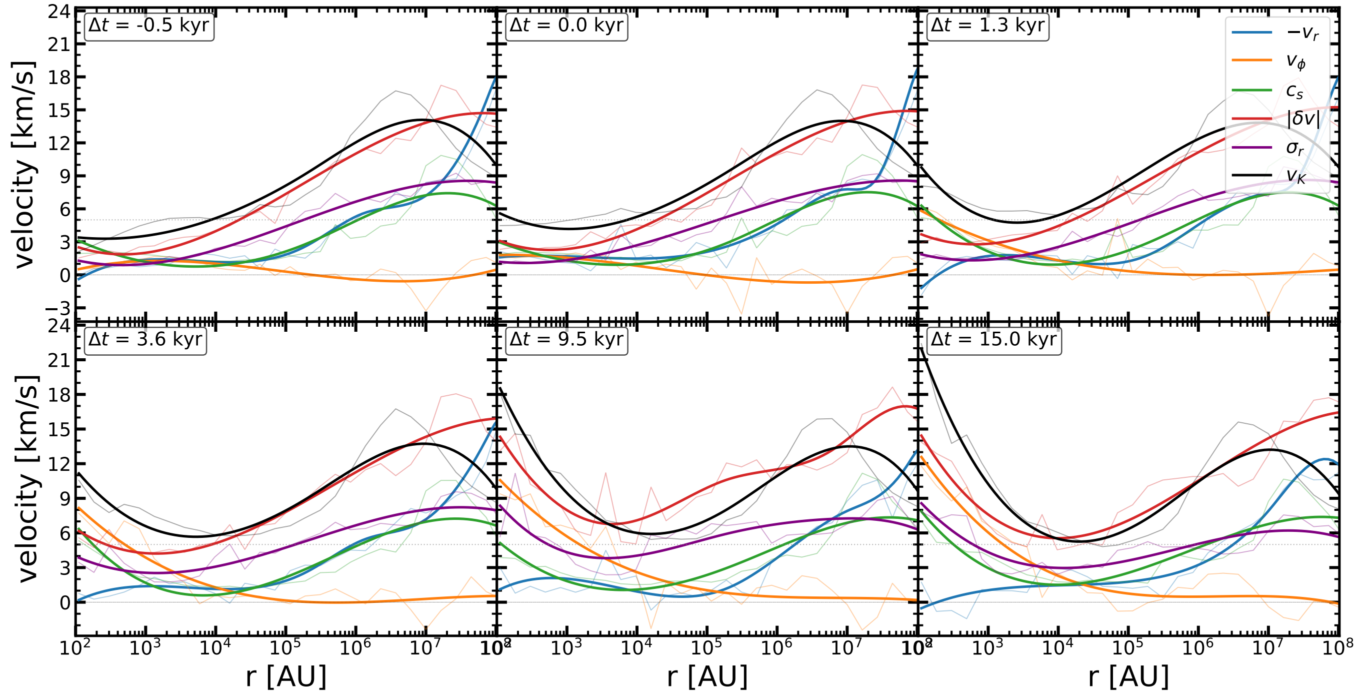

FIG. 7.— Same as ??, but for spherical shells, with radii ranging from inside of the disk ($r \leq 2500$ AU) to outside of the disk. Turbulence is supersonic at all times. Radial velocity is averaged out outside of disk (most of angular momentum is in the disk), gas is infalling at at the 100 pc ($10^7 - -10^8$ AU) scales.

For the local Toomre Q calculation we replace c_s with $\zeta = \sqrt{\sigma_r^2 + c_s^2}$, which includes both thermal and turbulent contributions, since supersonic turbulence provides additional effective pressure support.

For each snapshot the gas is binned into 20 annular bins in the face-on frame. In each annulus b we compute mass-weighted quantities:

$$\Sigma_b = \frac{\sum_{i \in b} m_i}{\pi(r_{\mathrm{hi},b}^2 - r_{\mathrm{lo},b}^2)}, \qquad \sigma_{r,b} = \sqrt{\frac{\sum_{i\in b} m_i\, v_{r,i}^2}{\sum_{i \in b} m_i} - \bar{v}_{r,b}^2} \tag{7}$$

The angular velocity is measured directly from the rotation profile:

$$\Omega_b = \frac{|\bar{v}_{\phi,b}|}{r_b} \tag{8}$$

where $\bar{v}_{\phi,b}$ is the mass-weighted mean azimuthal velocity in annulus b and r_b is the bin center. The epicyclic frequency is computed numerically from the measured rotation curve via (Pringle and King 2007).

$$\kappa_b^2 = \frac{2\,\Omega_b}{r_b} \left.\frac{\mathrm{d}(r^2\,\Omega)}{\mathrm{d}r}\right|_{r_b} \tag{9}$$

The Toomre parameter per annulus is then

$$Q_b = \frac{\zeta_b\,\kappa_b}{\pi\,G\,\Sigma_b} \tag{10}$$

A pixel-by-pixel face-on Q map is produced by computing local mass-weighted quantites for all of the relevant parameters:

$$Q(x,y) = \frac{\zeta_b(x,y)\;\kappa(r_{x,y})}{\pi\,G\,\Sigma(x,y)} \tag{11}$$

where $r_{x,y} = \sqrt{x^2+y^2}$, and $\zeta = \sqrt{\sigma_r^2 + c_s^2}$, which sums over thermal and turbulent contributions.

The Toomre Q parameter measures the gravitational stability of the disk. Figure 9 (top) shows face-on maps of the local Q, computed using the streaming-subtracted velocity dispersion σ_v in place of c_s. At the pre-stellar epoch ($t - t_1 = -0.5$ kyr), the gas is broadly unstable with $Q \lesssim 1$ throughout most of the region. By $t - t_1 = 0$, the central region near the first sink begins to stabilise, while the surroundings remain marginally unstable. At $t - t_1 \approx 1.3$ kyr, a clear boundary emerges between a stabilised inner region and an unstable outer envelope. By the time spiral arms have developed ($t - t_1 \approx 3.6$ kyr), a ring of $Q < 1$ material traces the spiral arms, while the inner disk is stabilised ($Q > 1$) by the strong velocity dispersion generated near the central sinks.

The radial Toomre Q profile (Figure 10) quantifies its evolution. At the pre-stellar epoch ($t - t_1 = -0.5$ kyr), Q is below unity at all radii ($Q \sim 0.2$–0.3), reflecting a globally unstable collapsing core. As the first sink forms and accretes, the inner Q rises: by $t - t_1 = 1.3$ kyr (green), $Q \approx 1$ throughout, consistent with a marginally stable state. By $t - t_1 = 9.5$ kyr (brown), Q is mostly stable in the whole disk, with central region being the most stable. At later times a radial gradient develops, with the inner disk more stable ($Q \gg 1$) and the outer spiral arms remaining near or below $Q \sim 1$.

Figure 9 (bottom) shows the space–time evolution of the azimuthally-averaged Toomre Q radial profile. Lines on the plot correspond to evolution of radial positions of sinks ($r(t)$).

The disk begins in a predominantly unstable state ($Q < 1$, red) at all radii. The first sink forms at the centre, and within ~ 1 kyr the $Q = 1$ contour (black line) begins to propagate outward from the centre, tracing a

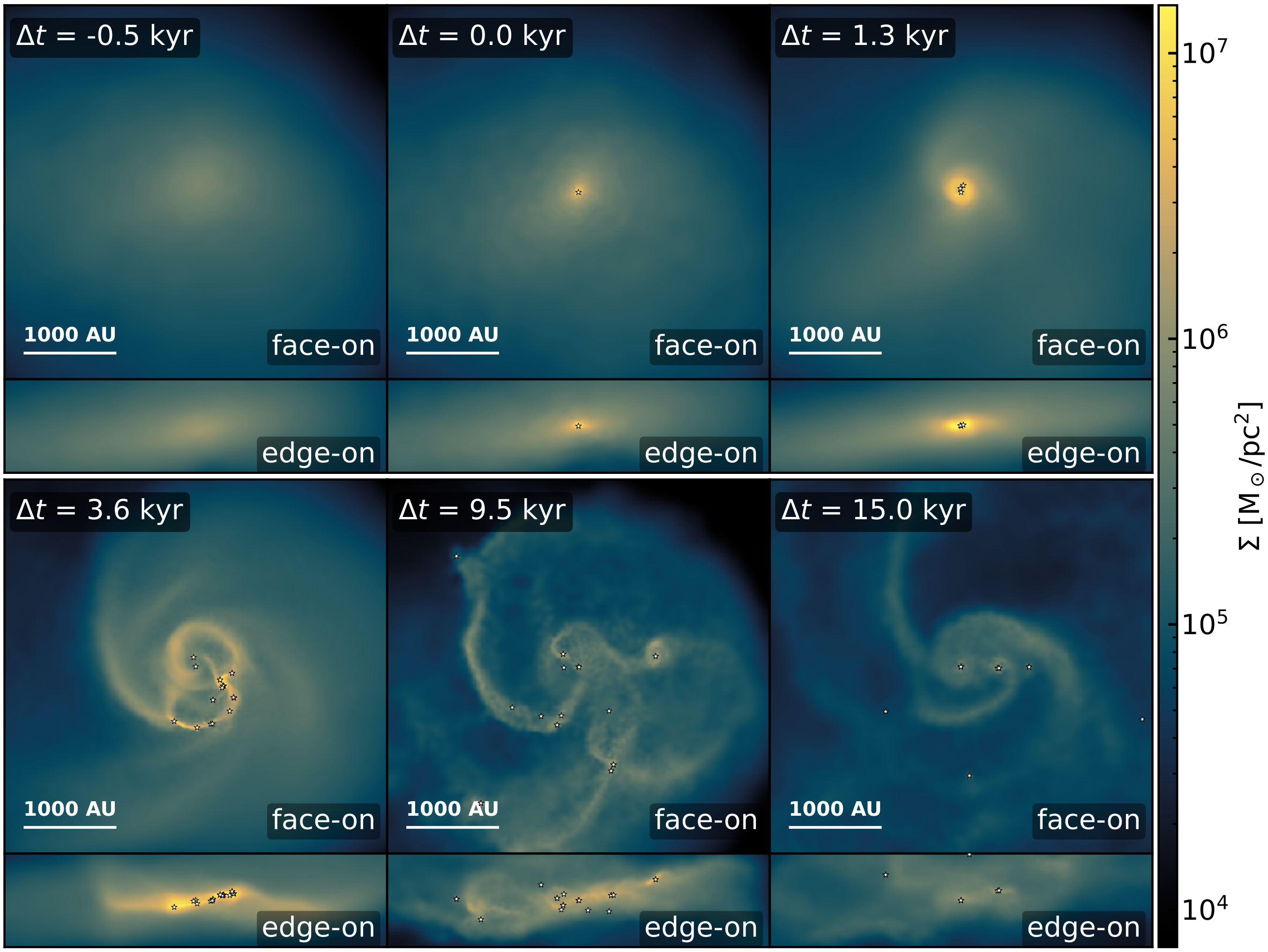

FIG. 8.— Face-on and edge-on surface density projections at six epochs. Each panel shows a 4100×4100 AU region centred on the most massive sink. Star symbols mark sink particles within the image box.

wave of stabilisation driven by the velocity dispersion generated during the initial collapse. A second stabilisation wave is launched by the formation of a second sink near the centre at $\Delta t \approx 1$ kyr. These waves propagate outward at speeds of ~ 2–$3\,c_s$.

After $t - t_1 \approx 3$ kyr, the $Q = 1$ boundary has swept through most of the disk. At this point, a burst of sink formation occurs, with sinks appearing at $r \sim 300$–800 AU clustered around $\Delta t \approx 3$–4 kyr. This burst takes out a lot of gas mass from the disk, stabilizing it. A lot of sinks also merge shortly after being produced in this event.

A similar burst, yet producing less sinks, happens later in time at around $\Delta t \approx 7$–8 kyr. It is preceded by "infalling region of instability" (patch with marginally higher Q values,) which disapperas for the same reasons as it does in the first bulk formation scenario. Much fewer sinks merge in this event, compared to the first one. Last star in the disk in our simulation forms in this event at around $\Delta t \approx 8$ kyr. All the sink formation history can also be seen on the number of sinks evolution plot in Figure 11, with clear jumps in number of sinks at $\Delta t \approx 3-4, 7-8$ kyr. 1 sink forms in the first kyr of the simulation, then only its companion forms as a stable second sink at 1 kyr, and the simulation has only 2 sinks for most of the time until onset of bulk star formation event at $\delta t \approx 3-4$ kyr. During that event, ≈ 20 sinks form and survive (with a few of them being absorbed into larger ones shortly after formation). Extra ≈ 8 non-merging sinks form during the second star formaion event during the second collapse of the spiral arm. Final number of sinks is 32,

3.3. *Magnetic fields*

Figures 12–13 show face-on ($z = 0$ slice) and edge-on ($x = 0$ slice) views of the toroidal and vertical components of the magnetic field at six epochs.

At the pre-stellar epoch, B_z shows large-scale coherent structure with a preferred sign, pointing towards primordial field orientation. As the disk forms and develops spiral structure, B_z is amplified in the disk midplane and develops sign reversals that trace the spiral arms, with field strengths reaching $|B_z| \sim 10^{-2}$–10^{-1} G at late times.

The toroidal component B_ϕ (Figure 12) develops an antisymmetric pattern by $t - t_1 \approx 1.3$ kyr: the field exhibits opposite signs on either side of the disk midplane, as confirmed by the edge-on view. This field can be achieved if we assume that primordial B-field was directed in +x +y direction diagonally. By $t - t_1 \approx 3.6$ kyr,

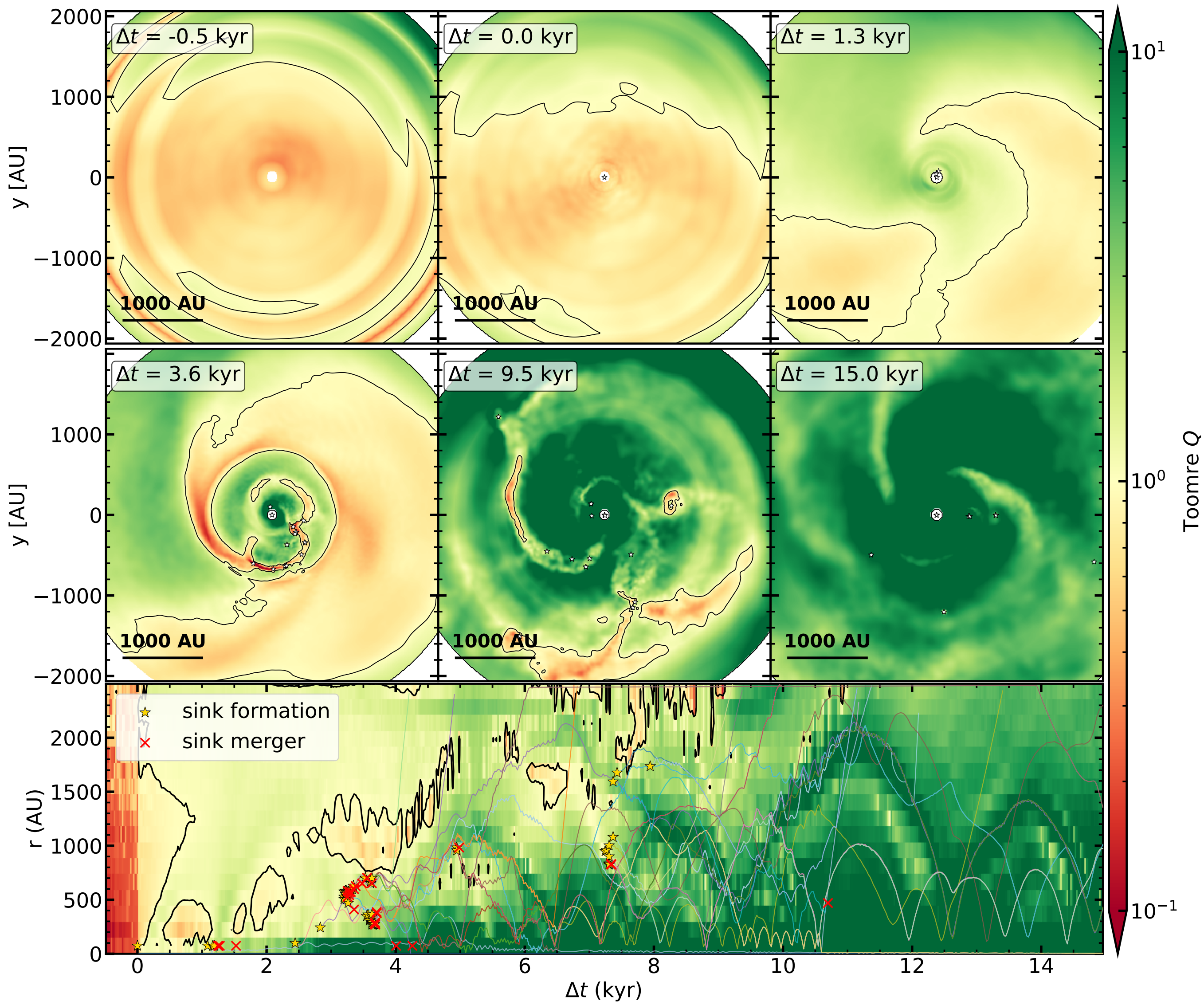

FIG. 9.— Top: Face-on maps of the local Toomre Q parameter at six epochs. The black contour marks $Q = 1$. Bottom: Evolution of radial profile of Toomre Q (heatmap), and sink formations, mergers, and radial positions' evolution (lines). At early times the gas is globally unstable ($Q < 1$); at late times the inner disk is strongly stabilised while the outer spiral arms remain near or below $Q = 1$. Radial profile of Toomre Q at six epochs. The horizontal dashed line marks $Q = 1$. At early times the entire region has $Q < 1$; at late times a radial gradient develops, with $Q > 10^3$ at $r < 100$ AU falling to $Q \sim 1$ at $r \sim 2000$ AU.

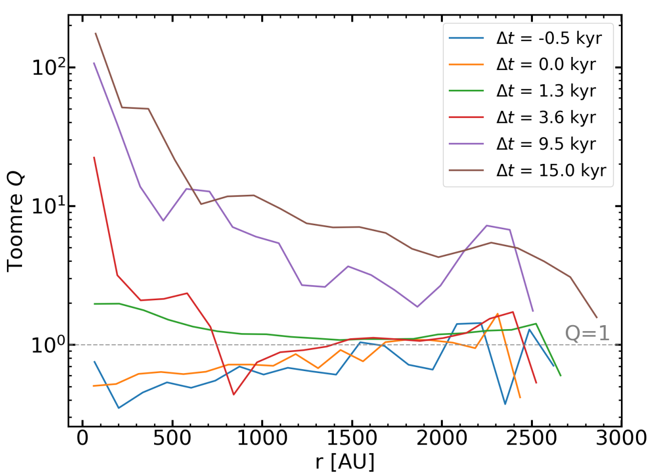

FIG. 10.— Toomre Q radial profiles time evolution. The disk starts

B_ϕ displays a clear two-armed spiral pattern with alternating positive and negative regions, reaching strengths of $|B_\phi| \sim 10^{-1}$ G. This antisymmetric, wound-up morphology is characteristic of field amplification by differential rotation and is consistent with a large-scale disk dynamo. By late times (15 kyr,) the toroidal component of the field is directed mostly opposite with respect to the direction of rotation of the disk.

Why did the toroidal field end up being anti-parallel to the direction of rotation? A clue might be located in the fourth face-on plot, corresponding to $\Delta t \approx 3.6$ kyr, roughly the time of first bulk star formation event. It is clear that at this state, the spacial distribution of the toroidal field is evenly distributed between positive ($B_{tor} > 0$, red) and negative ($B_{tor} < 0$, blue) components, with each of two of spiral arms corresponding to one of them. However, the spiral arm with denser gas corresponds to $B_{tor} < 0$ regime, and the magnetic field is

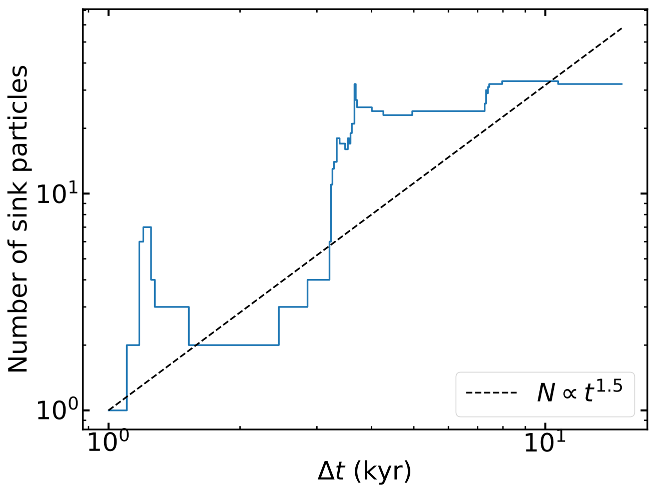

FIG. 11.— Log–log plot of the time evolution of the number of sinks in the simulation. The dashed line shows a $N \propto t^{1.5}$ power law for comparison (Li et al. 2018). Bulk star formation events in the spiral arms (visible in the bottom panels of Figure 9) are seen to rapidly increase the number of sinks at $t - t_1 \approx$ 3–4 kyr and 7–8 kyr.

amplified in there during collapse via flux-freezing mechanism. At later times this amplification is dispersed arond the whole disk region, making $B_{tor} < 0$ regime dominant.

We define mass-to-flux ration per annulus b as:

$$\mu_{\Phi,b} = \frac{2\pi\sqrt{G}\,\Sigma_b}{|B_{z,b}|} \tag{12}$$

The mass-to-flux ratio μ_Φ (Figure 15) evolves significantly over time.

At early times ($t - t_1 \leq 1.3$ kyr), $\mu_\Phi \sim$ 100–300 at all radii, indicating an extremely magnetically supercritical state in which gravity overwhelms magnetic support by more than two orders of magnitude. The mass-to-flux ratio then declines steadily as the field is amplified by compression and winding: by $t - t_1 = 3.6$ kyr (red), μ_Φ has dropped to $\sim$ 10–100. By $t - t_1 = 6.3$ kyr (purple), μ_Φ exhibits large radial scatter, with values ranging from ~ 1 to ~ 100 and dipping below the critical value ($\mu_\Phi < 1$) at some inner radii. At the latest epoch ($t - t_1 = 9.5$ kyr, brown), significant portions of the inner disk ($r \sim$ 200–500 AU) have $\mu_\Phi < 1$, indicating that the disk has become locally *magnetically subcritical* – a state where magnetic pressure and tension can in principle support the gas against gravitational collapse.

This transition from $\mu_\Phi \sim 300$ to $\mu_\Phi \lesssim 1$ over ~ 10 kyr represents a decrease of more than two orders of magnitude and implies a corresponding amplification of the magnetic field by flux-freezing compression, differential rotation winding, and possibly the turbulent dynamo.

Figure 14 shows evolution of phase plot of magnetic field magnitude $|B|$ through the 6 epochs we have been discussing. There is an evolving power law relationship that evolves as $|B| \propto n^{3/4} \rightarrow |B| \propto n^2$.

This shows that B-field amplification via dynamo is properly resolved in our simulation. but even with this **[TODO: More explanations needed. Is this true?]**

3.4. *Star Formation and the IMF*

Figure 16 shows the time evolution of gas, stellar (sink), and total mass within a 0.1 pc aperture centred on the disk.

The total stellar mass (orange, all sinks combined) grows from $\sim 1\,M_\odot$ at $t - t_1 = 1$ kyr to $\sim 200\,M_\odot$ by 15 kyr, with a best-fit power law $M_* \propto (t - t_0)^{2.08}$ (dashed red line). This is consistent with the $M_* \propto t^2$ scaling predicted by models of accelerating accretion from a turbulent collapsing core (McKee and Tan 2003). Murray et al. (2017) confirmed this scaling in high-resolution AMR simulations of self-gravitating turbulent collapse, finding that the density inside the stellar sphere of influence evolves to a time-independent attractor $\rho(r,t) \rightarrow \rho(r) \propto r^{-3/2}$, which, combined with Keplerian infall velocities, yields a mass accretion rate $\dot{M} \propto t$ and thus $M_* \propto t^2$. Our measured exponent of 2.08 extends the Murray et al. (2017) result from parsec-scale molecular cloud collapse to the sub-AU scales of Population III accretion disks.

The disk gas mass (cyan solid) starts at $\sim 50\,M_\odot$ and remains roughly constant for the first ~ 3 kyr, as infall from the envelope replenishes gas consumed by sink accretion. At later times, the disk gas mass begins to decline as the star formation efficiency rises. The gas mass within the 0.1 pc aperture (cyan dashed) is approximately constant at $\sim 200\,M_\odot$ throughout the simulation, yet its evolution follows one for the disk gas mass up until $\Delta \approx 11$ kyr, after which we see an increase in aperture gas mass, while disk gas mass stays constant. The reason for this is probably a large amount of infalling gas suddenly entering the aperture, resulting in a positive $\dot{M}_{gas,apert}$ (i.e. more gas is entering the system than is being converted to stellar mass). This sudden rise in the gas mass also "ignites" the stellar accretion again after its slow-down, driven by the depletion of initial accessible reservoirs in the original disk, and one can see this "bounce-back" in Figure 18. In addition to the increased accretion rates, we also see that the gas depletion rate ($-\dot{M}_{gas,apert}$) is not on the plot, which is the case since the total amount of gas is increasing, thus the gas depletion rate becoming negative right at the time of the "bounce-back".

The lower panel of Figure 16 shows the star formation efficiency (SFE) $f_* = M_*/(M_{\rm gas} + M_*)$. Within the disk alone, f_* rises steeply from $\sim 2\%$ at 1 kyr to $\sim 80\%$ by 10 kyr. Within the 0.1 pc aperture, f_* reaches $\sim 50\%$ by 10 kyr. These are high efficiencies compared to present-day star formation, but it is also not unexpected, since we did not pick an object, similar to a $z = 0$ giant molecular cloud (GMC) to compare our results with. Even our aperture will have less gas mass than present-day GMC, thus resulting in higher SFE.

Figure 18 shows the corresponding mass rates. The stellar accretion rate (magenta) fluctuates around $\sim 5 \times 10^{-3}\,M_\odot\,{\rm yr}^{-1}$ at early times, rising to $\sim 10^{-2}\,M_\odot\,{\rm yr}^{-1}$ at late times with strong variability. These rates are consistent with the canonical Population III accretion rate $\dot{M} \sim c_s^3/G \sim 10^{-3}\,M_\odot\,{\rm yr}^{-1}$ set by the sound speed in $\sim$ 200–1000 K gas (Abel et al. 2002; Stacy et al. 2010). The lower panel shows the sphere of influence radius $r_{\rm SOI}$, defined as the radius where the enclosed gas mass equals the total stellar mass ($M_{\rm gas,enc} = M_*$). This grows from

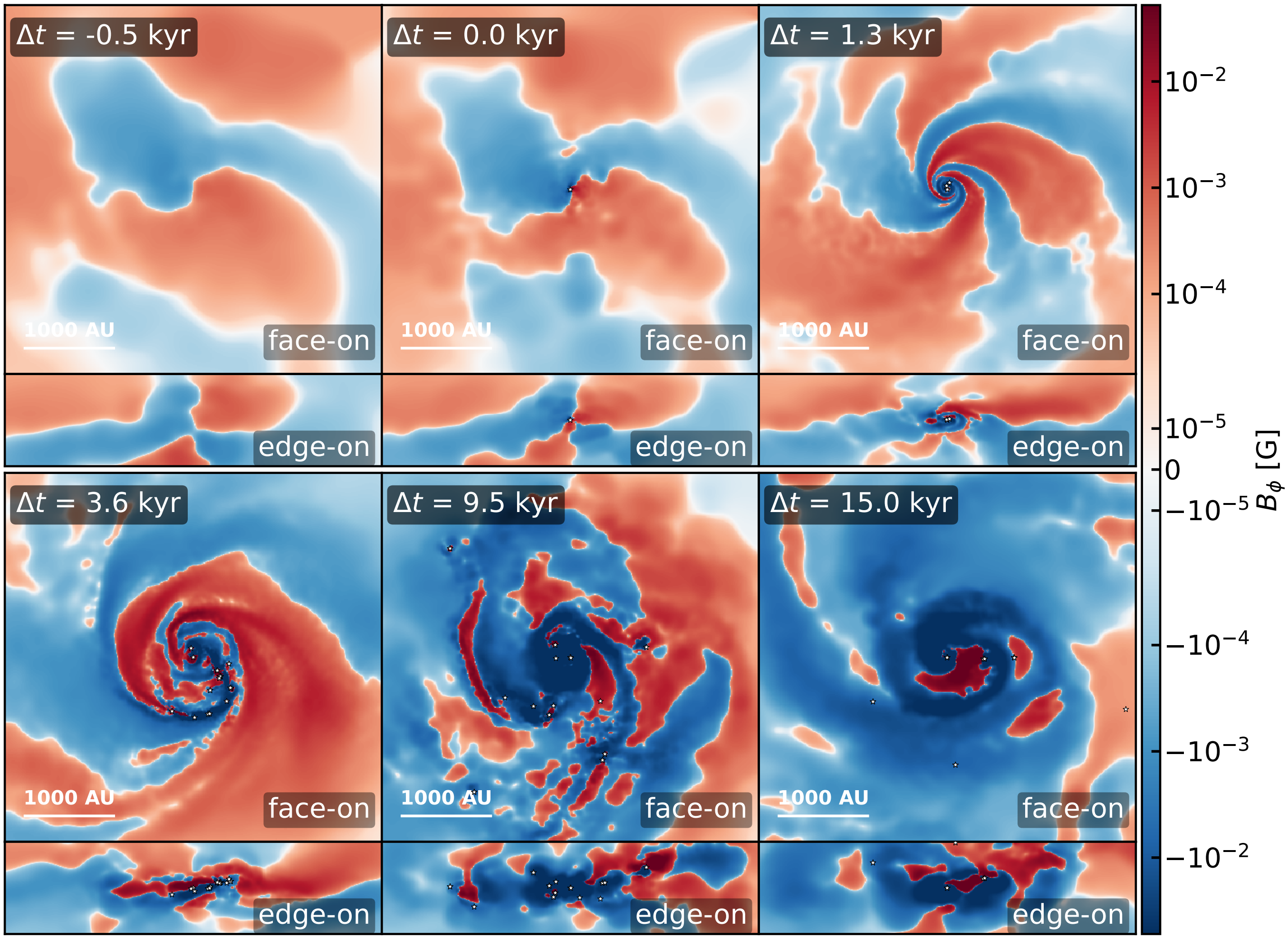

FIG. 12.— Face-on ($z = 0$ slice) and edge-on ($x = 0$ slice) of the toroidal magnetic field component B_ϕ at six epochs.

~ 80 AU at 1 kyr to $\sim 10^4$ AU by 10 kyr. It drops slightly with the new gas entering the aperture.

3.4.1. *Initial mass function*

Figure 19 shows the initial mass function (IMF) of the sinks at $t - t_1 = 15$ kyr, compared to the present-day Kroupa (2001) and Chabrier (2003) initial mass functions (IMFs). The high-mass slope of the IMF is well fitted by a power law $\mathrm{d}N/\mathrm{d}M \propto M^\alpha$ with $\alpha = -0.99^{+0.14}_{-0.14}$, shallower than Salpeter slope ($\alpha = -2.35$) and the Kroupa high-mass slope ($\alpha = -2.3$).

Figure 17 shows integrated accretion history of individual sinks. We can clearly see that 2 of the heaviest sinks are the first 2 sinks that formed in the simulation, and are located in the center of the disk. A lot of low-mass ($< 3M_\odot$) sinks stop accreting relatively early because they get ejected out of the disk, thus limiting the amount of gas, available to them. In our simulation, we have seen 14 out of 32 present sinks being ejected out of the disk by the 15 kyr time. With 62 total sink particles that were observed during the whole simulation time, the fraction of $\sim 52\%$ sinks being merged and 23% of sinks being ejected from the disk is consistent with values, expeced in the literature (compared to 50% and 25% respectively of (Stacy and Bromm 2013), or 30% of ejections of (Wollenberg et al. 2020)).

At the same figure it can be seen that the accretion in the center of the disk (heaviest 2 sinks) mostly halts, which is a consequence of depletion of gas reservoir. This gas depletion is shown an drop in stelllar accretion rate in Figure 18. The “bounce-back”, described in detail in subsection 3.4, seems to mostly target the non-central sinks, since we still observe their growth after $\Delta t \approx 11$ kyr. This competitive accretion in the cluster environment may preferentially feed the most massive objects (Bonnell et al. 2001), flattening the IMF slope.

4. COMPARISON TO PREVIOUS POPULATION III SIMULATIONS

Our simulation combines three features that, taken together, are uncommon in the existing literature: fully cosmological initial conditions, radiation-MHD with resolved individual protostellar formation (STARFORGE), and 15 kyr of post-collapse evolution in a halo progenitor of a Milky Way-mass system. Below we compare our results in each dimension to prior work.

4.1. *Fragmentation and sink counts*

The ~ 60 sink formation events we observe over 15 kyr place our simulation at the high end of the published range. Early simulations with cosmological initial conditions found 2–5 fragments per halo within the first few hundred years (Greif et al. 2011; Clark et al. 2011), while

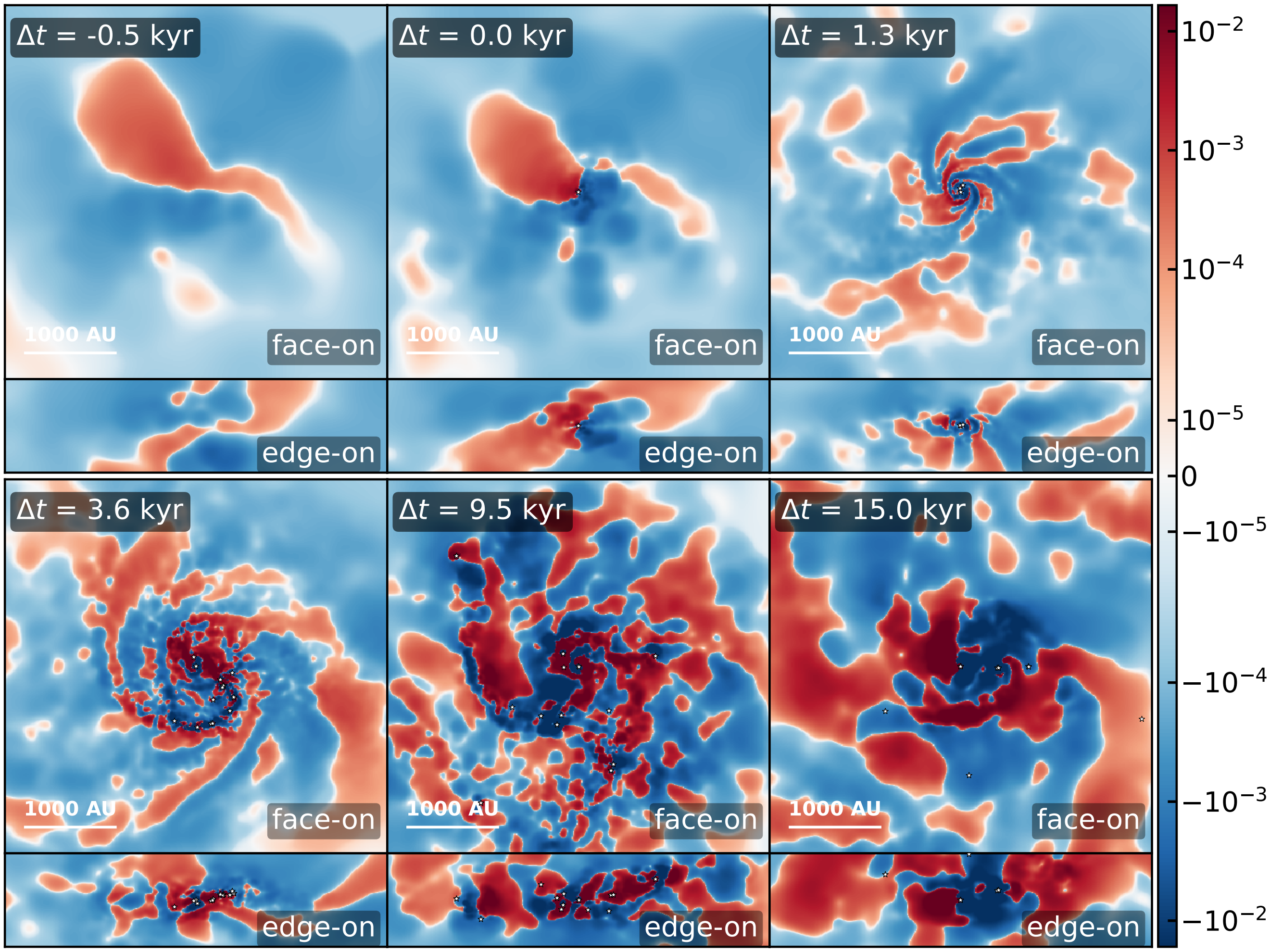

FIG. 13.— Face-on ($z = 0$ slice) and edge-on ($x = 0$ slice) of the vertical magnetic field component B_z at six epochs.

longer integrations over $\sim 5\,\mathrm{kyr}$ recovered ~ 10 sinks (Stacy et al. 2012). The radiation-hydrodynamic cosmological zoom-ins of Latif and Khochfar (2022) across five minihaloes produced up to 23 stars per halo (55 total), with masses 0.1–37 $M_\odot$ and a slope $\mathrm{d}N/\mathrm{d}M \propto M^{-1.18}$. The systematic study of Wollenberg et al. (2020) with 45 controlled simulations (varying rotation and turbulence) found 5–20 sinks per halo, confirming that fragmentation is highly stochastic and sensitive to initial conditions.

Our fragment count likely reflects three compounding factors. First, our 15 kyr integration is substantially longer than most published runs, allowing many cycles of instability, collapse, and re-accretion. Second, cosmological initial conditions from the m12f progenitor provide asymmetric, turbulent infall that continually replenishes the outer disk and seeds new unstable structures. Third, our mass-to-flux ratio remains comfortably supercritical ($\mu_\Phi \gg 1$) for most of the simulation, so magnetic fields do not suppress fragmentation here as they do in some studies (Stacy et al. 2022; Machida et al. 2025). The high fragment count is not unprecedented: Latif and Khochfar (2022) obtained comparable numbers with RHD (no MHD), and Prole et al. (2022) showed that sink counts increase monotonically with resolution without converging, suggesting our result may be a lower bound. Of the 62 total sink formation events, 32 survive to 15 kyr; the remainder were lost to mergers ($\sim 52\%$ of all events) or ejected from the disk ($\sim 23\%$), consistent with the rates reported by Stacy et al. (2012) and Wollenberg et al. (2020).

4.2. *Role of the large-scale environment*

A key distinction between our simulation and most previous work is the identity of the host halo. At the epoch of first collapse ($z \sim 21$), the m12f progenitor is a $\sim 10^5$–$10^6\,M_\odot$ minihalo – comparable in mass to those studied in most other work – but it is embedded in a large-scale overdensity destined to grow to $M_{\rm vir} \sim 10^{12}\,M_\odot$ by $z = 0$. The most directly comparable simulation, Meziani et al. (2026), targets the m09 halo whose progenitor at $z \sim 14$ resides in a quieter environment (reaching $\sim 2 \times 10^9\,M_\odot$ at $z = 0$), and finds a *single* Pop III protostar growing to $\sim 27\,M_\odot$ over 31 kyr – in stark contrast to our ~ 60-fragment cluster. Although this comparison is a single pair of halos, the contrast suggests that large-scale environment may be a key driver of whether Population III star formation produces a single massive object or a protostellar cluster. The overdense filamentary infall geometry of the m12f progenitor naturally seeds higher angular momentum, more complex disk structure, and stronger turbulence than the more isolated m09 environment. Chen et al. (2025) showed that infall

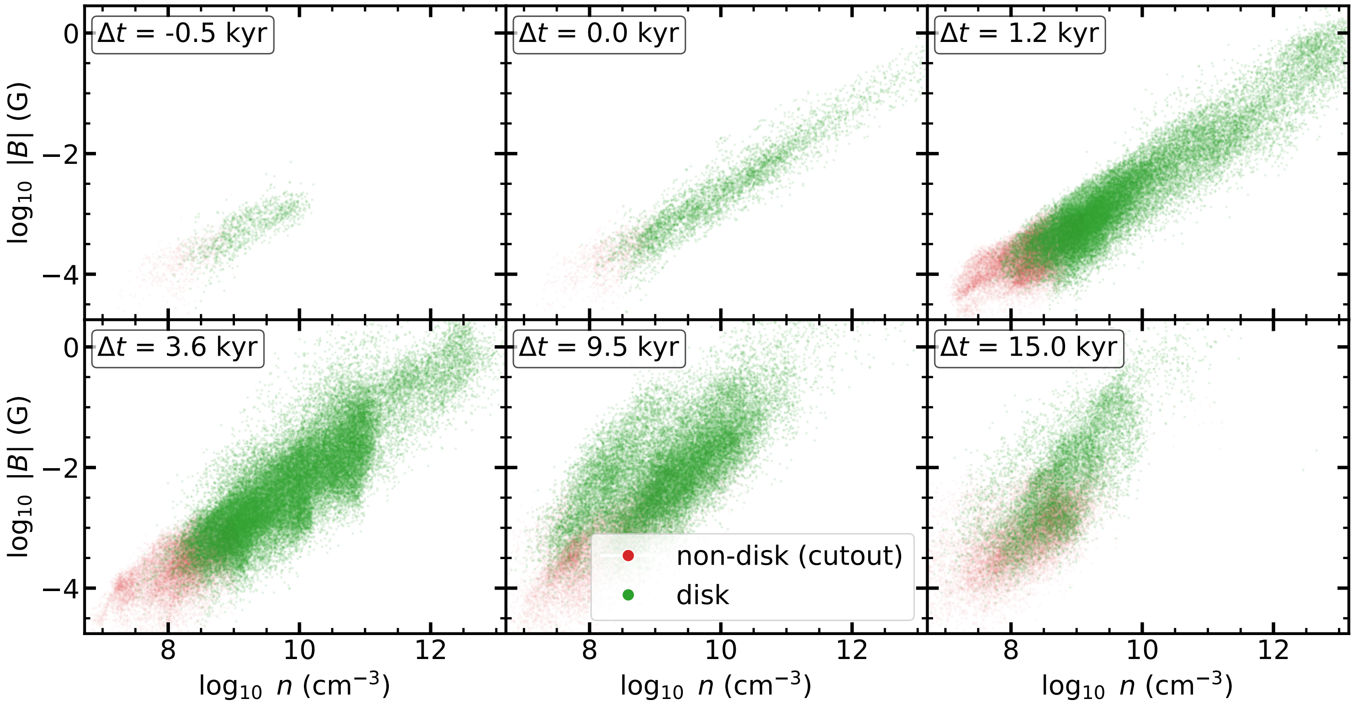

FIG. 14.— Evolution of phase plot of magnitude of magnetic field $|B|$ vs number density n. Green points correspond to particles, identified as inside of the disk, and red points correspond to particles in the cutout simulation that are outside of the disk. $|B|$ spans 3 orders of magnitude at the point of first sink formation, and up to 4 orders of magnitude at later times.

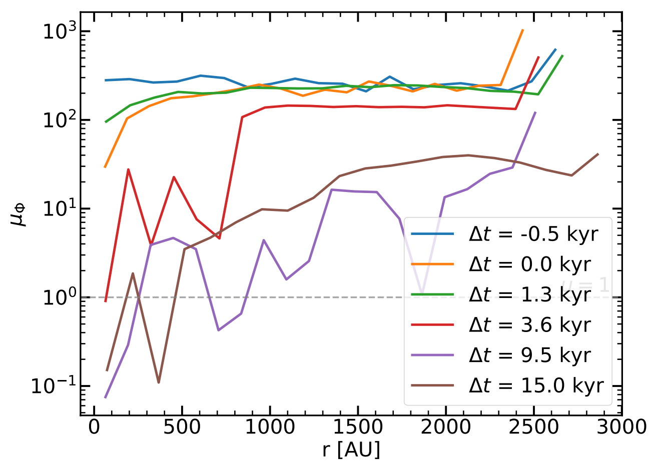

FIG. 15.— Dimensionless mass-to-flux ratio μ_Φ as a function of radius at six epochs. The dashed line marks the critical value $\mu_\Phi = 1$. The disk evolves from highly supercritical ($\mu_\Phi \sim 100$–300) at early times to partially subcritical ($\mu_\Phi < 1$) at some inner radii by 9.5 kyr.

into dark-matter potential wells drives supersonic turbulence ($\mathcal{M} \sim 2$–4) across all minihaloes; the m12f overdensity likely places our halo toward the high-$\mathcal{M}$ end of this distribution. Testing this environmental dependence statistically will require a suite of cosmological RMHD zoom-ins spanning a range of $z = 0$ halo masses. The kinematic structure of our simulation supports this picture: the radial velocity v_r is subsonic within the disk but transitions to supersonic outside it, indicating that the outer envelope is in a state of continuous supersonic infall that feeds the disk from above.

4.3. *Stellar mass growth and accretion rates*

The total stellar mass of $\sim 200\,M_\odot$ at 15 kyr is broadly consistent with canonical Population III accretion rates. Stacy et al. (2010) found $\dot{M} \sim 10^{-3}\,M_\odot\,\mathrm{yr}^{-1}$ sustained over $\sim$ 5 kyr; integrating this rate over 15 kyr gives $\sim 15\,M_\odot$, far less than we find, reflecting our larger sink count sharing a common reservoir rather than a single accreting object. Our time-averaged rate of $\sim 10^{-2}\,M_\odot\,\mathrm{yr}^{-1}$ (across all sinks) is at the high end of the range 10^{-4}–$10^{-2}\,M_\odot\,\mathrm{yr}^{-1}$ found by Hirano et al. (2014) across 100 minihaloes, again consistent with a particularly turbulent, high-infall-rate environment.

The power-law growth $M_* \propto t^{2.08}$ confirms the $M_* \propto t^2$ scaling found by Murray et al. (2017) in simulations of self-gravitating turbulent collapse, where the density inside the sphere of influence converges to $\rho \propto r^{-3/2}$ and Keplerian infall gives $\dot{M} \propto t$. Our slightly super-quadratic exponent likely reflects the additional contribution of continued cosmological infall sustaining the gas reservoir at large radii, absent in the isolated box setup of Murray et al. (2017). The POPSICLE simulations of Sharda and Menon (2025) produce a single most-massive protostar of $\sim 65\,M_\odot$ within 5 kyr in their RMHD run; our total mass of $\sim 200\,M_\odot$ is distributed across ~ 30 surviving sinks at 15 kyr, implying a characteristic fragment mass of ~ 5–$7\,M_\odot$, consistent with fragmentation-induced starvation redistributing mass from the most massive object to the cluster population (Prole et al. 2022). Sugimura et al. (2020) and Sugimura et al. (2023) produced massive Population III binaries with individual masses $> 30\,M_\odot$ at separations > 2000 AU; the two most massive sinks in our simulation – which dominate the mass budget throughout – may represent an analogous central binary, with the surrounding cluster population arising from subsequent disk fragmentation.

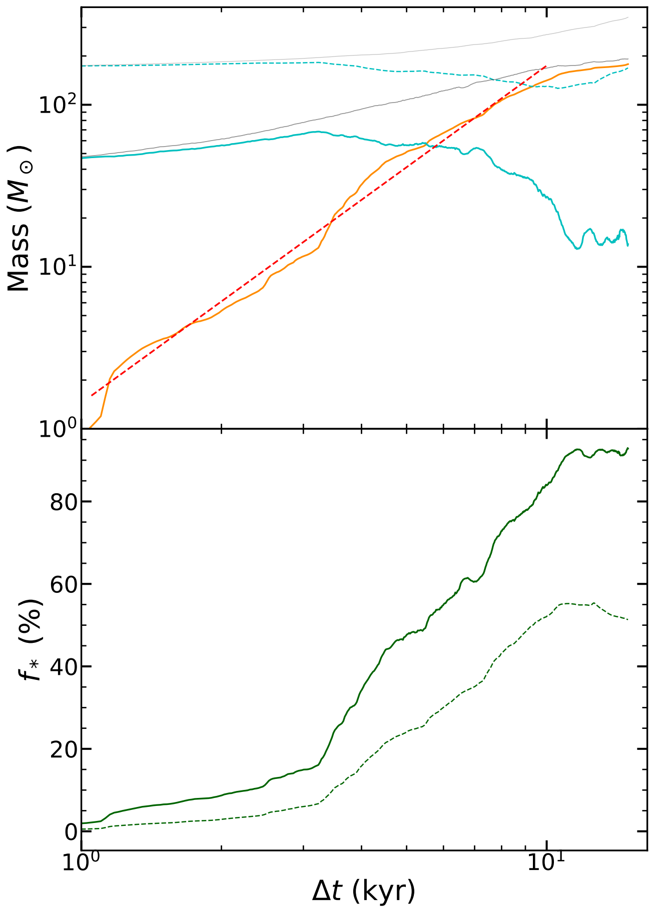

FIG. 16.— Top: time evolution of the stellar mass (all sinks; orange), disk gas mass (cyan solid), gas within a 0.1 pc aperture (cyan dashed), and total mass (disk + stars, grey solid; aperture + stars, grey). The best-fit power law $M_* \propto t^{2.08}$ is shown as a dashed red line; the reference t^2 scaling as a dotted pink line. Bottom: star formation efficiency f_* within the disk (green solid) and within the 0.1 pc aperture (green dashed).

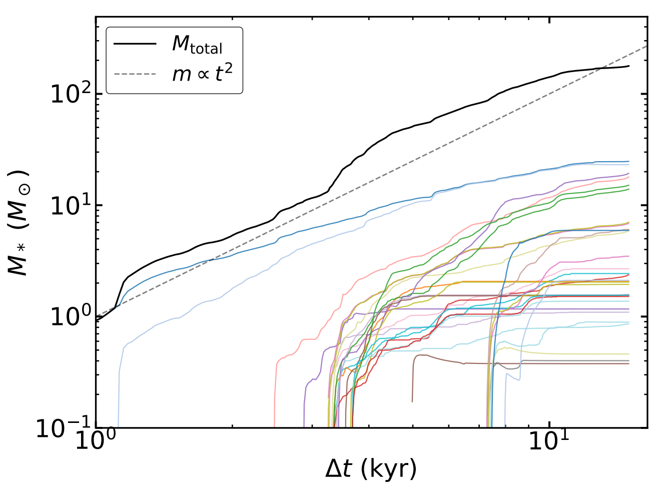

FIG. 17.— Evolution of masses of individual sinks. Dashed line represents ideal $M_* \propto t^2$ law, which is being followed for total mass of all the sinks.

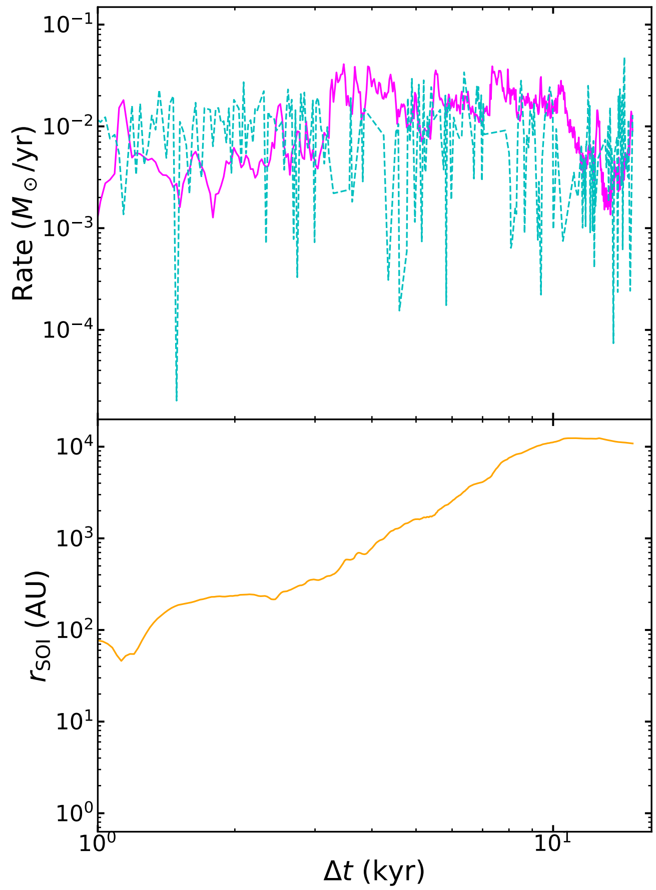

FIG. 18.— Top: stellar accretion rate (magenta) and gas depletion rate within the aperture (cyan dashed). Bottom: sphere of influence radius $r_{\rm SOI}$ ($M_{\rm gas,enc} = M_*$; orange), growing from ~ 80 AU to $\sim 10^4$ AU over the simulation.

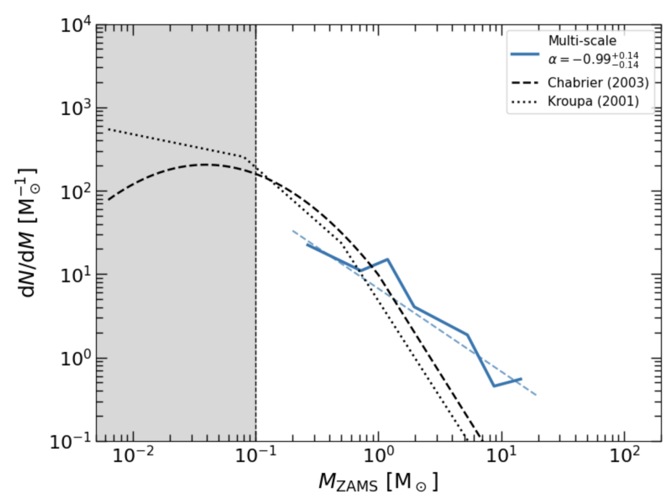

FIG. 19.— Protostellar mass function at $t - t_1 = 15$ kyr (blue solid histogram) compared to the Chabrier (2003) (dashed) and Kroupa (2001) (dotted) present-day IMFs. The best-fit high-mass power-law slope is $\alpha = -0.99^{+0.14}_{-0.14}$ (light blue dashed line), shallower than but comparable to the Salpeter slope (-2.35). The grey shaded region marks masses below the completeness limit of the simulation. As this is a initial mass function measured at 15 kyr, the final masses will be modified by continued accretion, mergers, ejections, and radiative feedback.

4.4. *Star formation efficiency*

The star formation efficiency we find ($f_* \sim 50$–80% within the disk at 15 kyr) is high compared to both present-day values ($\sim$ 1–10%) and to other Population III simulations. Hirano et al. (2023) report $\epsilon_{\rm III} \sim 10^{-2}$ in their large halo sample at the point of initial collapse; our higher integrated value reflects the longer post-collapse integration time and the lack of negative feedback mechanisms that would quench accretion. Radiative feedback is included in our simulation (M1 radiation transport), but within the 15 kyr of evolution covered here no protostar has yet grown sufficiently massive and contracted sufficiently to launch an ionising HII region that would reverse the accretion flow. Hosokawa et al. (2011) found that UV feedback halts accretion at $\sim 43\,M_\odot$ for a single protostar on a $\sim 10^5$ yr Kelvin-Helmholtz timescale; with our mass distributed across many objects, no individual protostar yet dominates the radiative output, and the global accretion efficiency remains high. This suggests that the high SFE we observe is a transient feature of the early accretion phase; longer integrations incorporating protostellar contraction and UV feedback would be required to determine final stellar masses and the termination of star formation. Protostellar jets are active in the simulation but have not yet launched strongly enough to disrupt the global accretion flow at 15 kyr: none of our protostars has yet contracted onto the main sequence, so the jet driving power remains limited. Jets are expected to become progressively more important at later times as individual protostars contract and their surface escape velocities increase, potentially quenching accretion and reducing the SFE from its current high value (Grudić et al. 2021; Meziani et al. 2026).

4.5. *Protostellar mass function*

Our protostellar mass function (PMF) at 15 kyr has a high-mass slope $\alpha = -0.99^{+0.14}_{-0.14}$, shallower than the Salpeter slope (-2.35) but broadly consistent with other Population III simulations that find top-heavy, log-flat distributions. Latif and Khochfar (2022) find $\alpha \approx -1.18$ across five halos with RHD; Stacy et al. (2016) and Greif et al. (2011) both report roughly log-flat mass functions extending from sub-solar to $\gtrsim 10\,M_\odot$. We emphasise that our PMF is a snapshot at 15 kyr, not a final IMF: all sinks are still actively accreting, no radiative feedback has terminated accretion, and the population will be further shaped by continued mergers, ejections, and eventual UV feedback (Hosokawa et al. 2011, 2016; Stacy et al. 2016). The two most massive sinks, both located at the center of the disk, have each reached $\sim 25\,M_\odot$ by 15 kyr – approaching the $\sim 43\,M_\odot$ UV-feedback threshold of Hosokawa et al. (2011) – with their accretion rates flattening due to local gas depletion in the inner disk. The outer sinks, by contrast, continue to grow by accreting from the surrounding envelope, so the cluster mass budget shifts progressively outward. The $\sim 52\%$ merger fraction and $\sim 23\%$ ejection fraction we observe are consistent with the $\sim 50\%$ and $\sim 25\%$ values of Stacy et al. (2012) and the $\sim 30\%$ ejection rate of Wollenberg et al. (2020). Latif and Khochfar (2022) report an unusually high ejection fraction of $\sim 70\%$, which they attribute to the strong three-body interactions driven by the radiative heating of the disk; the lower fraction in our simulation may reflect the slower build-up of radiation pressure in a multi-sink cluster where no single object yet dominates.

4.6. *Magnetic field evolution*

Our mass-to-flux ratio evolves from $\mu_\Phi \sim 100$–300 at early times to $\mu_\Phi \lesssim 1$ in the inner disk by 10 kyr, with magnetic energy growing by ~ 4 orders of magnitude. This evolution is more dramatic than the $\mu_\Phi \sim 10$–100 reported by Machida et al. (2008) in idealised MHD simulations, likely because our longer integration time allows more flux accumulation by differential rotation winding and compression. Higashi et al. (2024) demonstrated that the turbulent small-scale dynamo amplifies fields to 30–50% of equipartition during the primordial collapse phase; our highly supercritical starting point and the continued amplification we observe are consistent with this picture.

Despite this amplification, disk fragmentation in our simulation remains vigorous throughout, in contrast to the strong suppression found in some MHD simulations. Machida et al. (2025) systematically varied initial field strength from $B_0 = 10^{-20}$ to 10^{-4} G and found that even the weakest field suppresses fragmentation and drives protostellar mergers within the first ~ 1000 yr – a timescale much shorter than our run. Stacy et al. (2022) found only two secondary sinks in cosmological MHD runs. The key difference is that our simulation starts from a highly supercritical configuration ($\mu_\Phi \sim 300$) and remains in the fragmentation-permitting regime ($\mu_\Phi > 1$) for the majority of the 15 kyr. Magnetic braking and flux-freezing act on a timescale long compared to the free-fall time at the disk scale, so fragmentation proceeds largely unimpeded while the field is amplified gradually from below. Only toward the end of our simulation does the inner disk approach the magnetically subcritical threshold; at this point, Meziani et al. (2026) find that protostellar jets are launched, a phase not yet reached in our 15 kyr window. The magnetic energy power spectrum is consistent with a Burgers spectrum ($P_B \propto k^{-2}$) at most times, oscillating between Burgers and Kolmogorov regimes but remaining closer to Burgers, indicating shock-dominated rather than purely subsonic turbulent amplification. A detailed analysis of the magnetic power spectrum and its implications for dynamo action in primordial disks is deferred to a companion paper. In contrast, Sadanari et al. (2021) and Sadanari et al. (2023) found that ambipolar diffusion is dynamically subdominant in primordial gas (diffusion heating always smaller than compressional heating), broadly consistent with our non-ideal MHD treatment where ideal MHD dominates throughout (Meziani et al. 2026). Tanaka et al. (2024) similarly found in primordial disk simulations that turbulent magnetic fields reduce but do not eliminate fragmentation, consistent with our result.

5. CONCLUSIONS

We have presented a radiation-magnetohydrodynamic cosmological zoom-in simulation of Population III star formation, following the collapse of primordial gas in the FIRE m12f halo from cosmological initial conditions through the formation and evolution of a protostellar accretion disk over ~ 15 kyr. Our main conclusions are:

1. **Vigorous disk fragmentation.** The accretion disk becomes gravitationally unstable within $\sim 3\,\mathrm{kyr}$ of the first protostar's formation, developing spiral arms that fragment into ~ 60 sink formation events over $15\,\mathrm{kyr}$, producing a total stellar mass of $> 150\,M_\odot$.

2. **Magnetic field amplification toward subcriticality.** The mass-to-flux ratio drops from $\mu_\Phi \sim 100$–300 at early times to $\mu_\Phi \lesssim 1$ at some inner radii by $10\,\mathrm{kyr}$, driven by flux-freezing, differential rotation winding, and possibly the turbulent dynamo. Magnetic energy grows by ~ 4 orders of magnitude. The inner disk eventually crosses the magnetically subcritical threshold.

3. **Sub-virial start to virial equilibrium.** The disk starts in a sub-virial state ($2E_{\mathrm{kin}}/|E_{\mathrm{pot}}| \approx 0.6$) and reaches virial equilibrium within $\sim 2\,\mathrm{kyr}$, thereafter self-regulating around $2E_{\mathrm{kin}}/|E_{\mathrm{pot}}| \approx 1$.

4. **Accelerating accretion and high SFE.** The stellar mass grows as $M_* \underset{\sim}{\propto} t^2$ at accretion rates of $\sim 10^{-3}$–$10^{-2}\,M_\odot\,\mathrm{yr}^{-1}$. The star formation efficiency reaches $\sim 80\%$ within the disk and $\sim 50\%$ within a 0.1 pc aperture.

5. **Top-heavy initial mass function.** The initial mass function has a high-mass slope $\alpha = -0.99^{+0.14}_{-0.14}$, shallower than the Salpeter/Kroupa value ($-2.35/-2.3$).

6. **Cosmological ICs matter.** The use of cosmological initial conditions produces asymmetric infall, complex angular momentum structure, and turbulent boundary conditions that promote vigorous fragmentation and a rich protostellar population, distinct from what is found in idealised setups. Continued accretion from LSS onto the protostellar disk modifies the IMF, flattening the slope.

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